Experiment 3: Email Spam or Ham Classification using Naive Bayes, KNN, and SVM

```
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```

1. Aim and Objectives

Aim

The aim of this experiment is to design and implement machine learning models for classifying emails as **spam** or **ham**. This experiment focuses on applying **Naïve Bayes**, **K-Nearest Neighbors (KNN)**, and **Support Vector Machine (SVM)** algorithms to understand how different classifiers and their variants perform on real-world email data. The experiment also aims to analyze model effectiveness using standard evaluation metrics and validation techniques.

Objectives

- To preprocess and analyze the email dataset using exploratory data analysis (EDA) techniques.
- To build and train Naïve Bayes, KNN, and SVM classifiers with different variants and parameter settings.
- To evaluate and compare model performance using accuracy, precision, recall, F1-score, ROC curve, and confusion matrix.
- To perform K-Fold cross-validation and identify the most effective classifier for email spam detection.

2 Libraries Used

In [119…

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    confusion_matrix, ConfusionMatrixDisplay,
    roc_curve, auc
)
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import classification_report
```

3 Code for All Variants and Models

3.1 1: Load and preprocess the dataset (missing values, normalization)

In [120…

```
import pandas as pd

# Load dataset
df = pd.read_csv("spambase_csv.csv")

# Show first 5 rows
df.head()
```

Out[120…

	word_freq_make	word_freq_address	word_freq_all	word_freq_3d	word_freq_our	word_f
0	0.00	0.64	0.64	0.0	0.32	
1	0.21	0.28	0.50	0.0	0.14	
2	0.06	0.00	0.71	0.0	1.23	
3	0.00	0.00	0.00	0.0	0.63	
4	0.00	0.00	0.00	0.0	0.63	

5 rows × 58 columns

In [121…

```
print("Shape:", df.shape)
df.info()
```

```
Shape: (4601, 58)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4601 entries, 0 to 4600
Data columns (total 58 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   word_freq_make              4601 non-null   float64
 1   word_freq_address           4601 non-null   float64
 2   word_freq_all               4601 non-null   float64
 3   word_freq_3d                4601 non-null   float64
 4   word_freq_our               4601 non-null   float64
 5   word_freq_over              4601 non-null   float64
 6   word_freq_remove            4601 non-null   float64
 7   word_freq_internet          4601 non-null   float64
 8   word_freq_order             4601 non-null   float64
 9   word_freq_mail              4601 non-null   float64
 10  word_freq_receive           4601 non-null   float64
 11  word_freq_will              4601 non-null   float64
 12  word_freq_people            4601 non-null   float64
 13  word_freq_report            4601 non-null   float64
 14  word_freq_addresses         4601 non-null   float64
 15  word_freq_free              4601 non-null   float64
 16  word_freq_business          4601 non-null   float64
 17  word_freq_email             4601 non-null   float64
 18  word_freq_you               4601 non-null   float64
 19  word_freq_credit            4601 non-null   float64
 20  word_freq_your              4601 non-null   float64
 21  word_freq_font              4601 non-null   float64
 22  word_freq_000               4601 non-null   float64
 23  word_freq_money             4601 non-null   float64
 24  word_freq_hp                4601 non-null   float64
 25  word_freq_hpl               4601 non-null   float64
 26  word_freq_george            4601 non-null   float64
 27  word_freq_650               4601 non-null   float64
 28  word_freq_lab               4601 non-null   float64
 29  word_freq_labs              4601 non-null   float64
 30  word_freq_telnet            4601 non-null   float64
 31  word_freq_857               4601 non-null   float64
 32  word_freq_data              4601 non-null   float64
 33  word_freq_415               4601 non-null   float64
 34  word_freq_85                4601 non-null   float64
 35  word_freq_technology        4601 non-null   float64
 36  word_freq_1999              4601 non-null   float64
 37  word_freq_parts             4601 non-null   float64
 38  word_freq_pm                4601 non-null   float64
 39  word_freq_direct            4601 non-null   float64
 40  word_freq_cs                4601 non-null   float64
 41  word_freq_meeting           4601 non-null   float64
 42  word_freq_original          4601 non-null   float64
 43  word_freq_project           4601 non-null   float64
 44  word_freq_re                4601 non-null   float64
 45  word_freq_edu               4601 non-null   float64
 46  word_freq_table             4601 non-null   float64
 47  word_freq_conference        4601 non-null   float64
 48  char_freq_%3B               4601 non-null   float64
 49  char_freq_%28               4601 non-null   float64
```

```
 50  char_freq_%5B               4601 non-null   float64
 51  char_freq_%21               4601 non-null   float64
 52  char_freq_%24               4601 non-null   float64
 53  char_freq_%23               4601 non-null   float64
 54  capital_run_length_average  4601 non-null   float64
 55  capital_run_length_longest  4601 non-null   int64
 56  capital_run_length_total    4601 non-null   int64
 57  class                       4601 non-null   int64
dtypes: float64(55), int64(3)
memory usage: 2.0 MB
```

In [122…

```
print("Missing values per column:")
print(df.isnull().sum())
```

```
Missing values per column:
word_freq_make                0
word_freq_address             0
word_freq_all                 0
word_freq_3d                  0
word_freq_our                 0
word_freq_over                0
word_freq_remove              0
word_freq_internet            0
word_freq_order               0
word_freq_mail                0
word_freq_receive             0
word_freq_will                0
word_freq_people              0
word_freq_report              0
word_freq_addresses           0
word_freq_free                0
word_freq_business            0
word_freq_email               0
word_freq_you                 0
word_freq_credit              0
word_freq_your                0
word_freq_font                0
word_freq_000                 0
word_freq_money               0
word_freq_hp                  0
word_freq_hpl                 0
word_freq_george              0
word_freq_650                 0
word_freq_lab                 0
word_freq_labs                0
word_freq_telnet              0
word_freq_857                 0
word_freq_data                0
word_freq_415                 0
word_freq_85                  0
word_freq_technology          0
word_freq_1999                0
word_freq_parts               0
word_freq_pm                  0
word_freq_direct              0
word_freq_cs                  0
word_freq_meeting             0
word_freq_original            0
word_freq_project             0
word_freq_re                  0
word_freq_edu                 0
word_freq_table               0
word_freq_conference          0
char_freq_%3B                 0
char_freq_%28                 0
char_freq_%5B                 0
char_freq_%21                 0
char_freq_%24                 0
char_freq_%23                 0
capital_run_length_average    0
```

```
capital_run_length_longest    0
capital_run_length_total      0
class                         0
dtype: int64
```

In [123…

```
X = df.drop('class', axis=1)
y = df['class']
```

In [124…

```
df.shape
```

Out[124…

```
(4601, 58)
```

In [125…

```
df.describe()
```

Out[125…

	word_freq_make	word_freq_address	word_freq_all	word_freq_3d	word_freq_our	w
count	4601.000000	4601.000000	4601.000000	4601.000000	4601.000000	
mean	0.104553	0.213015	0.280656	0.065425	0.312223	
std	0.305358	1.290575	0.504143	1.395151	0.672513	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	0.000000	0.000000	0.000000	
50%	0.000000	0.000000	0.000000	0.000000	0.000000	
75%	0.000000	0.000000	0.420000	0.000000	0.380000	
max	4.540000	14.280000	5.100000	42.810000	10.000000	

8 rows × 58 columns

3.2 Exploratory Data Analysis (EDA)

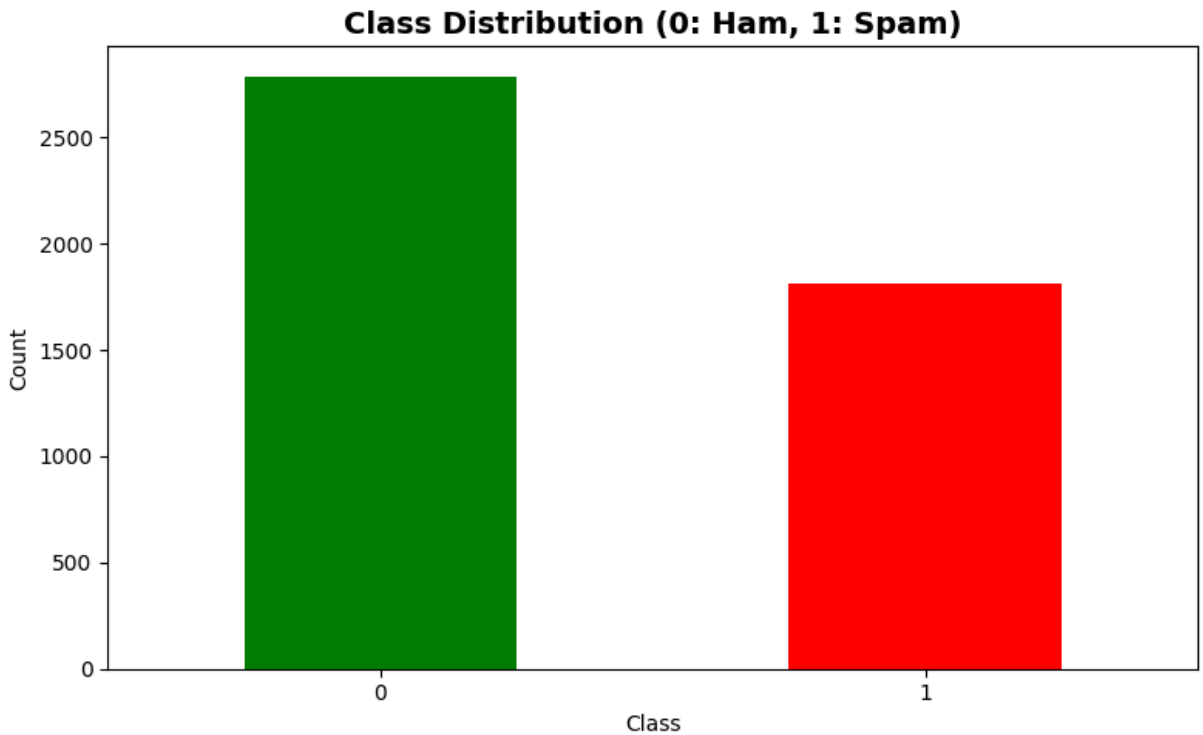
In [142…

```
print("\nClass Distribution:")
print(df.iloc[:, -1].value_counts())
```

```
Class Distribution:
class
0    2788
1    1813
Name: count, dtype: int64
```

In [143…

```
plt.figure(figsize=(8, 5))
df.iloc[:, -1].value_counts().plot(kind='bar', color=['green', 'red'])
plt.title('Class Distribution (0: Ham, 1: Spam)', fontsize=14, fontweight='bold')
plt.xlabel('Class')
plt.ylabel('Count')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



In [144…

```
class_counts = df.iloc[:, -1].value_counts()
print(f"\nClass Balance:")
print(f"Ham (0): {class_counts[0]/len(df)*100:.2f}%")
print(f"Spam (1): {class_counts[1]/len(df)*100:.2f}%")
```

```
Class Balance:
Ham (0): 60.60%
Spam (1): 39.40%
```

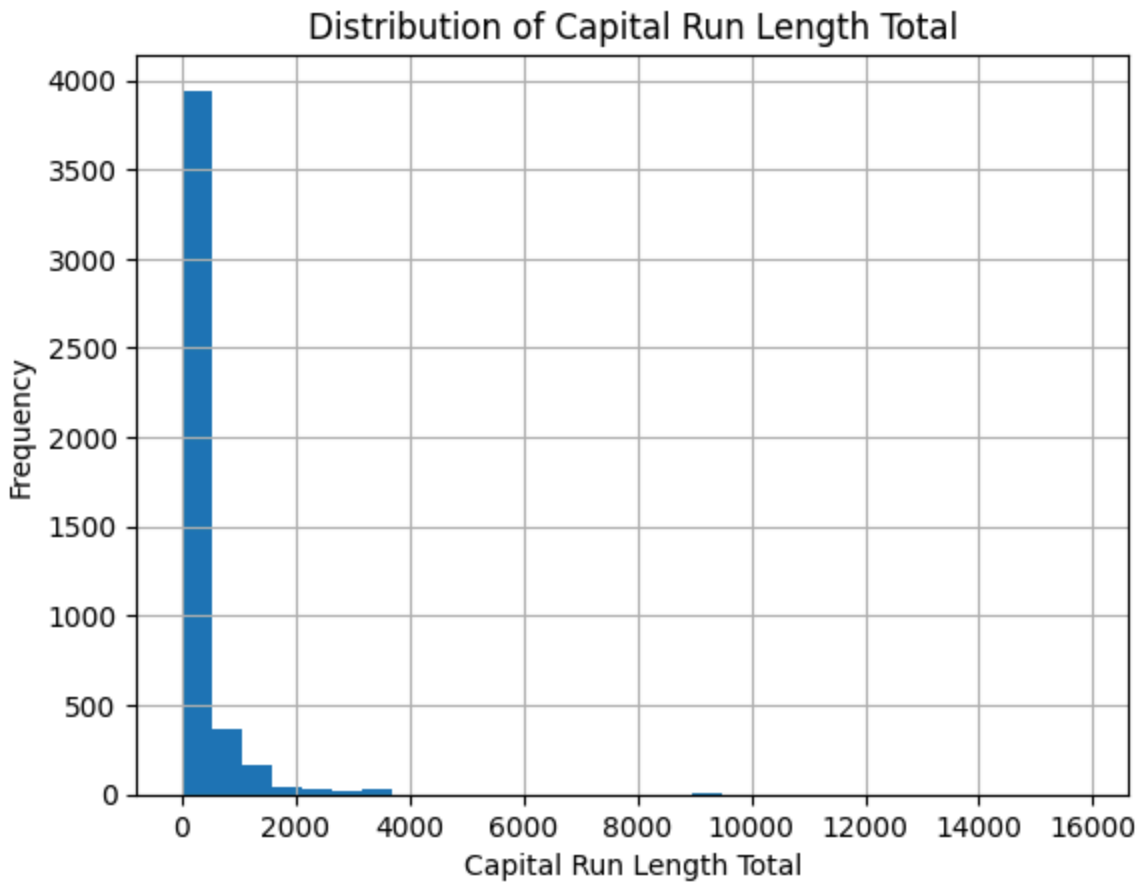
In [145…

```
features=df.columns[:-1]
len(features)
```

Out[145… 57

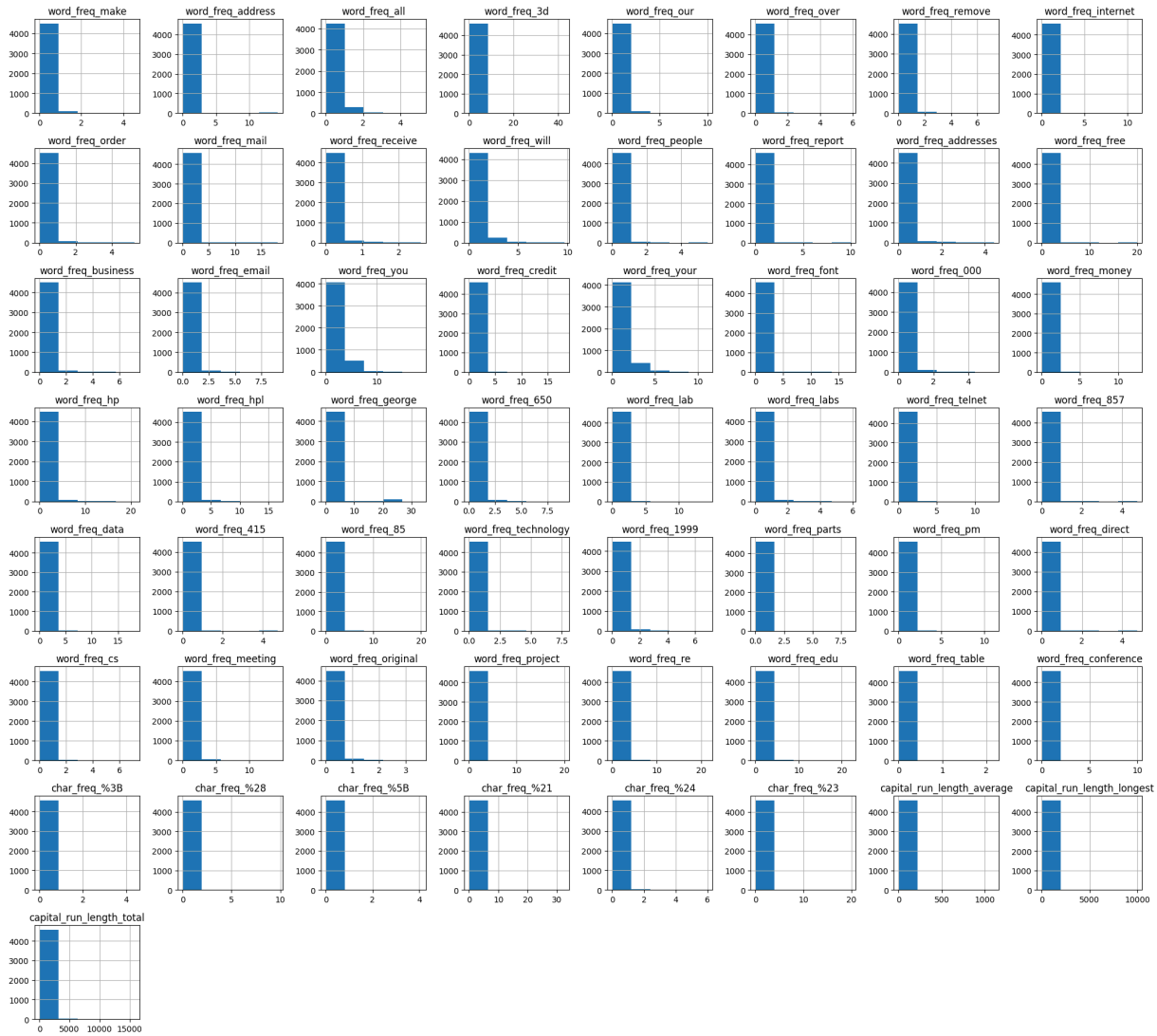
In [128…

```
plt.figure()
df['capital_run_length_total'].hist(bins=30)
plt.xlabel("Capital Run Length Total")
plt.ylabel("Frequency")
plt.title("Distribution of Capital Run Length Total")
plt.show()
```



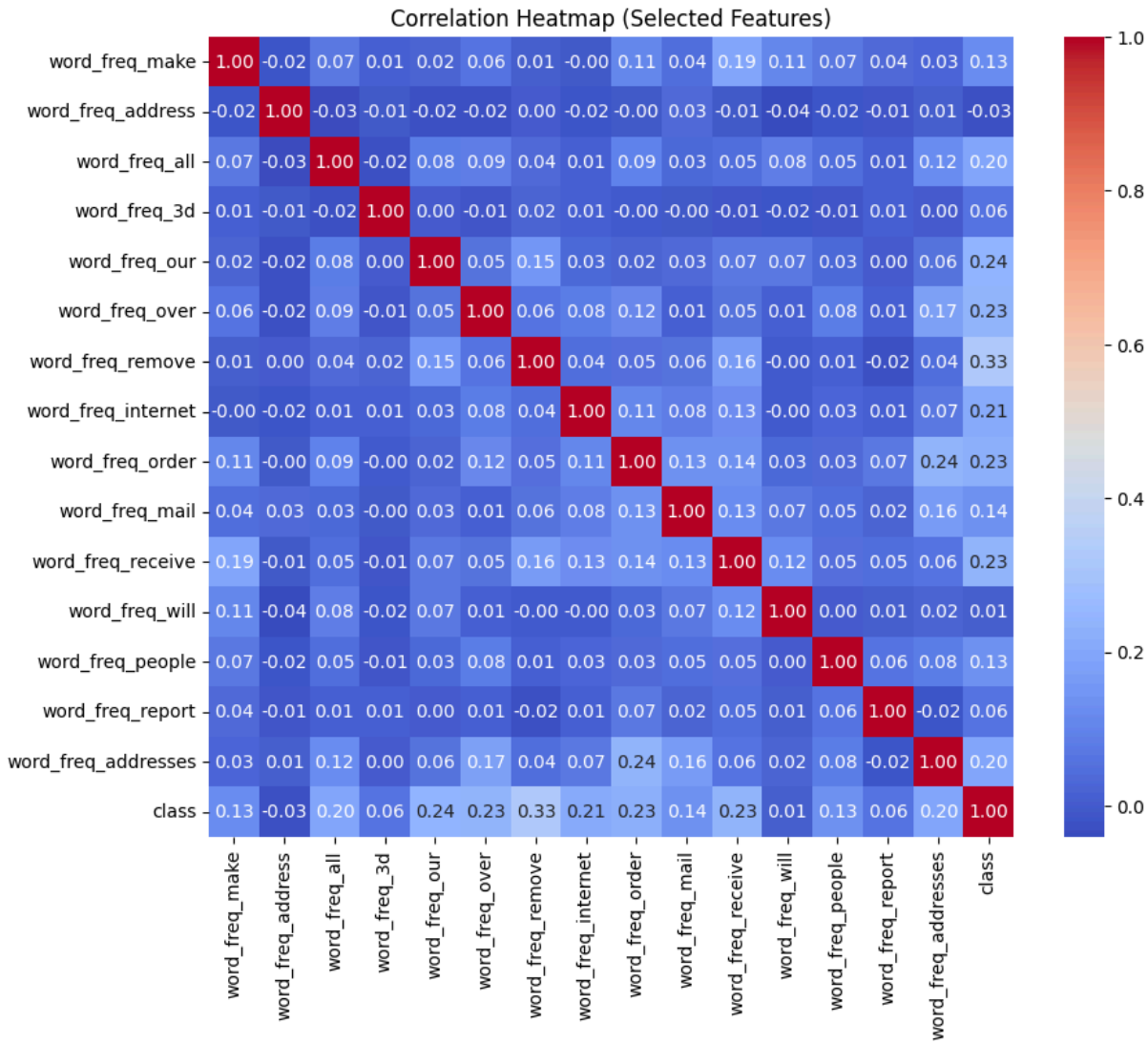
In [147…

```
df[features].hist(bins=5,figsize=(20,18))
plt.tight_layout()
plt.show()
```



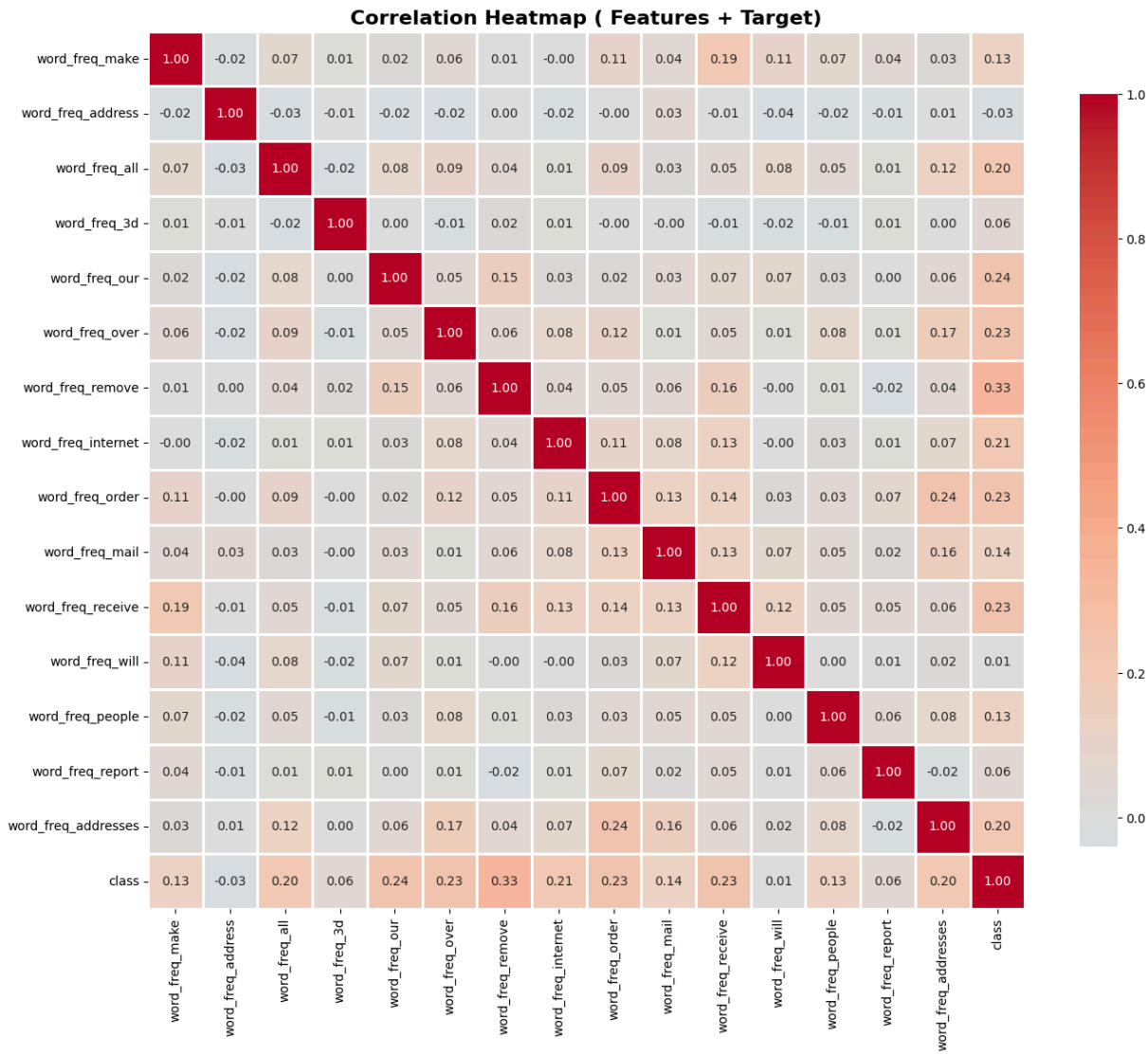
In [129…

```
# Select a subset of features for clarity
selected_features = list(X.columns[:15]) + ['class']

corr_matrix = df[selected_features].corr()

plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt='.2f')
plt.title("Correlation Heatmap (Selected Features)")
plt.show()
```



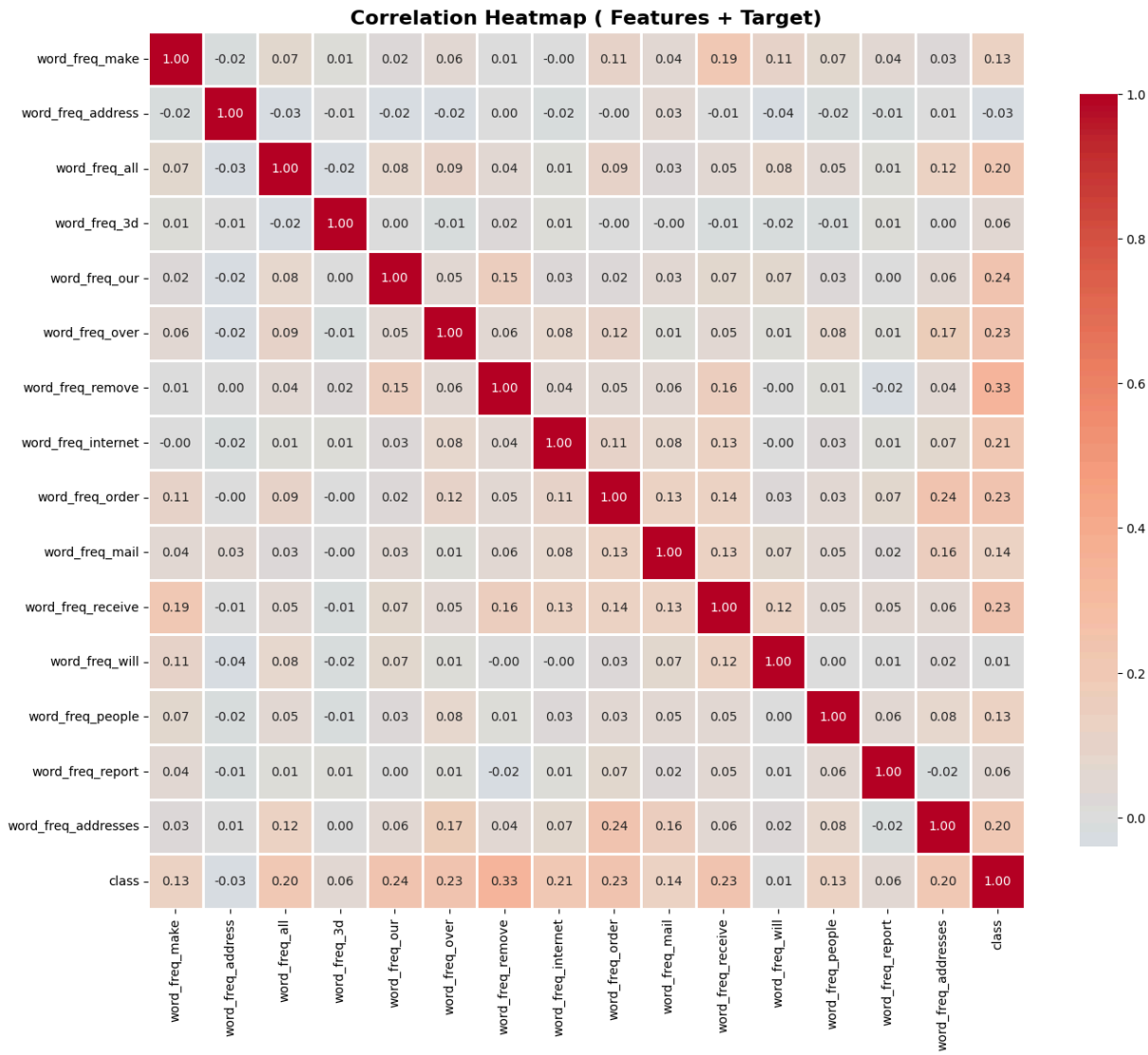
In [130…

```
correlation_matrix = df[selected_features].corr()

plt.figure(figsize=(14, 12))
sns.heatmap(correlation_matrix, annot=True, fmt='.2f', cmap='coolwarm',
            center=0, square=True, linewidths=1, cbar_kws={"shrink": 0.8})
plt.title('Correlation Heatmap ( Features + Target)',
          fontsize=16, fontweight='bold')
plt.tight_layout()
plt.show()
```



In [131…

```
import seaborn as sns
plt.figure(figsize=(14, 12))
sns.heatmap(correlation_matrix, annot=True, fmt='.2f', cmap='coolwarm',
            center=0, square=True, linewidths=1, cbar_kws={"shrink": 0.8})
plt.title('Correlation Heatmap ( Features + Target)',
          fontsize=16, fontweight='bold')
plt.tight_layout()
plt.show()
```



In [148…

```
corr_label=df.corr()["class"].sort_values(ascending=False)
corr_label
```

Out[148…

```
class                          1.000000
word_freq_your                 0.383234
word_freq_000                  0.334787
word_freq_remove               0.332117
char_freq_%24                  0.323629
word_freq_you                  0.273651
word_freq_free                 0.263215
word_freq_business             0.263204
capital_run_length_total       0.249164
word_freq_our                  0.241920
char_freq_%21                  0.241888
word_freq_receive              0.234529
word_freq_over                 0.232604
word_freq_order                0.231551
word_freq_money                0.216111
capital_run_length_longest     0.216097
word_freq_internet             0.206808
word_freq_email                0.204208
word_freq_all                  0.196988
word_freq_addresses            0.195902
word_freq_credit               0.189761
word_freq_mail                 0.138962
word_freq_people               0.132927
word_freq_make                 0.126208
capital_run_length_average     0.109999
word_freq_font                 0.091860
char_freq_%23                  0.065067
word_freq_report               0.060027
word_freq_3d                   0.057371
word_freq_will                 0.007741
word_freq_address             -0.030224
word_freq_parts               -0.031035
word_freq_table               -0.044679
char_freq_%3B                 -0.059630
char_freq_%5B                 -0.064709
word_freq_direct              -0.064801
word_freq_conference          -0.084020
char_freq_%28                 -0.089672
word_freq_project             -0.094594
word_freq_cs                  -0.097375
word_freq_415                 -0.112754
word_freq_857                 -0.114214
word_freq_data                -0.119931
word_freq_pm                  -0.122831
word_freq_telnet              -0.126912
word_freq_lab                 -0.133523
word_freq_original            -0.135664
word_freq_technology          -0.136134
word_freq_meeting             -0.136615
word_freq_re                  -0.140408
word_freq_edu                 -0.146138
word_freq_85                  -0.149225
word_freq_650                 -0.158800
word_freq_labs                -0.171095
word_freq_1999                -0.178045
word_freq_george              -0.183404
```

```
word_freq_hpl                -0.232968
word_freq_hp                 -0.256723
Name: class, dtype: float64
```

3.3 3:Split into train and test sets.

In [150…

```
x = df.iloc[:, :-1].values
y = df.iloc[:, -1].values
```

In [151…

```
print(f"\nFeatures shape: {X.shape}")
print(f"Target shape: {y.shape}")
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.3, random_state=42, stratify=y
)
print(f"\nTraining set size: {X_train.shape[0]}")
print(f"Test set size: {X_test.shape[0]}")
```

```
Features shape: (4601, 57)
Target shape: (4601,)

Training set size: 3220
Test set size: 1381
```

In [152…

```
from sklearn.preprocessing import MinMaxScaler
scaler =MinMaxScaler()
X_train_scaled=scaler.fit_transform(X_train)
X_test_scaled=scaler.transform(X_test)
```

In [154…

```
def evaluate_model(y_true, y_pred, model_name):
    accuracy = accuracy_score(y_true, y_pred)
    precision = precision_score(y_true, y_pred)
    recall = recall_score(y_true, y_pred)
    f1 = f1_score(y_true, y_pred)
    print(f"\n{'='*60}")
    print(f"{model_name} - Performance Metrics")
    print(f"{'='*60}")
    print(f"Accuracy: {accuracy:.4f}")
    print(f"Precision: {precision:.4f}")
    print(f"Recall: {recall:.4f}")
    print(f"F1-Score: {f1:.4f}")
    return {
    'accuracy': accuracy,
    'precision': precision,
    'recall': recall,
    'f1_score': f1
    }


def plot_confusion_matrix(y_true, y_pred, title):
    cm = confusion_matrix(y_true, y_pred)
    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
        xticklabels=['Ham', 'Spam'],
        yticklabels=['Ham', 'Spam'])
    plt.title(f'Confusion Matrix - {title}', fontsize=14, fontweight='bold')
```

```
    plt.ylabel('Actual')
    plt.xlabel('Predicted')
    plt.tight_layout()
    plt.show()
    return cm
def plot_roc_curve(y_true, y_pred_proba, title):
    fpr, tpr, _ = roc_curve(y_true, y_pred_proba)
    roc_auc = auc(fpr, tpr)
    plt.figure(figsize=(8, 6))
    plt.plot(fpr, tpr, color='darkorange', lw=2,
        label=f'ROC curve (AUC = {roc_auc:.4f})')
    plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--',
        label='Random Classifier')
    plt.xlim([0.0, 1.0])
    plt.ylim([0.0, 1.05])
    plt.xlabel('False Positive Rate')
    plt.ylabel('True Positive Rate')
    plt.title(f'ROC Curve - {title}', fontsize=14, fontweight='bold')
    plt.legend(loc="lower right")
    plt.grid(alpha=0.3)
    plt.tight_layout()
    plt.show()
    return roc_auc
```

3.4 4: Training the Models

3.4.1Naive Bayes Variant

1.GaussianNB

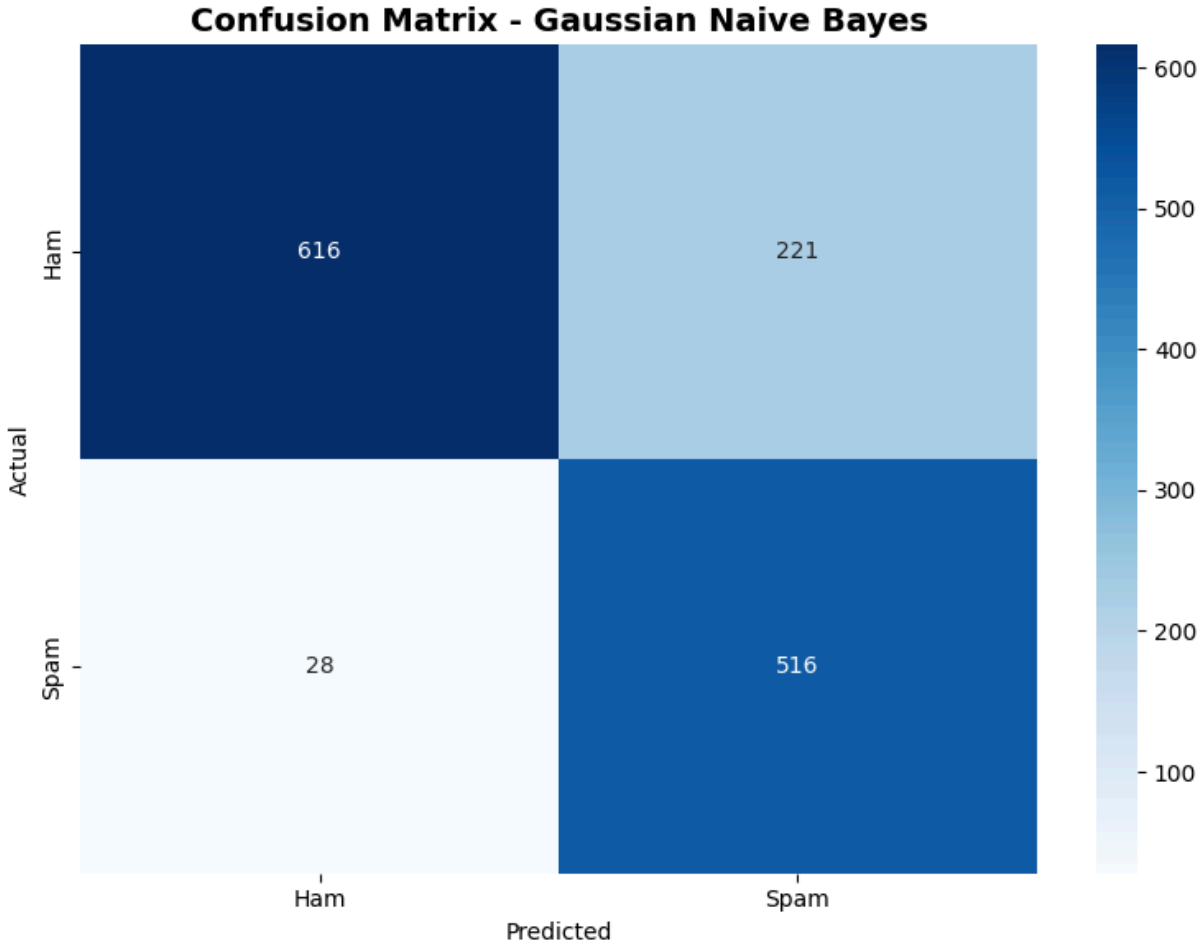
In [155…

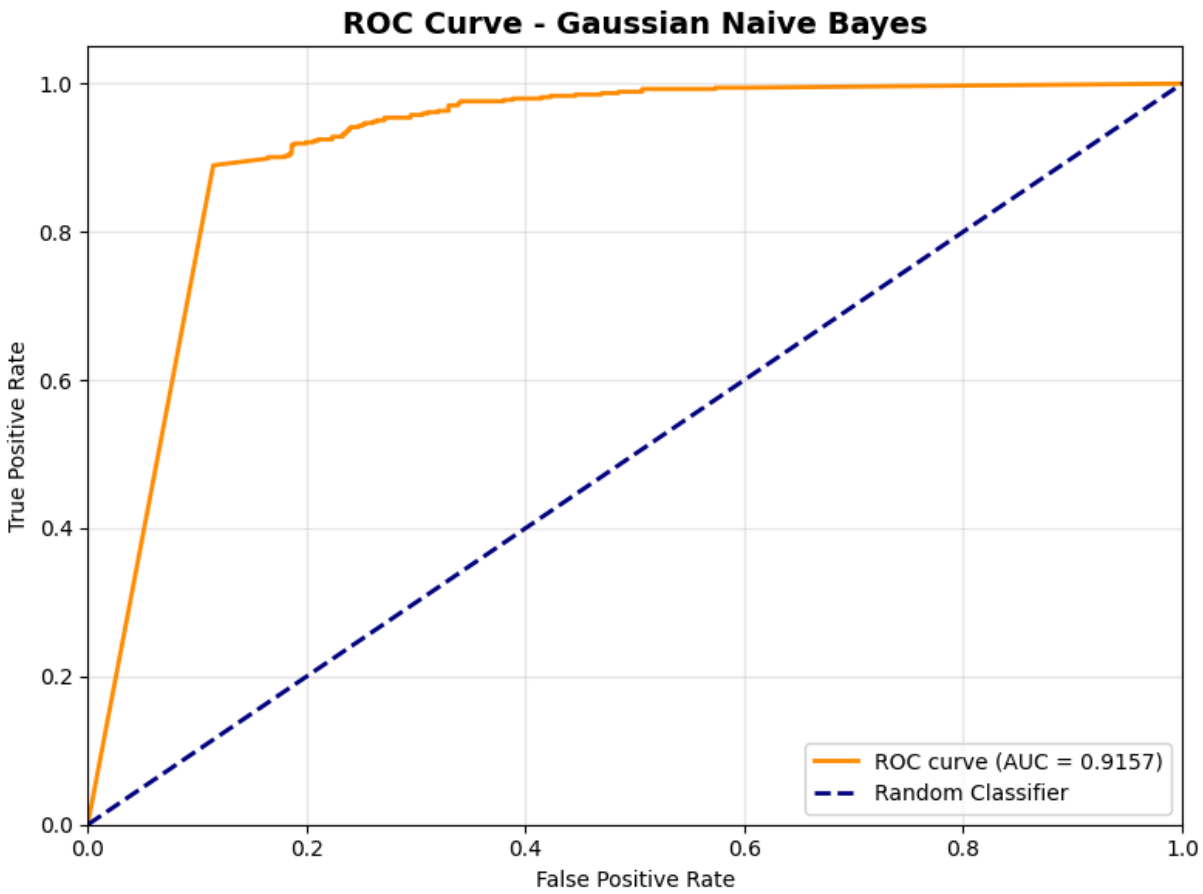
```
gnb = GaussianNB()
gnb.fit(X_train_scaled, y_train)
y_pred_gnb = gnb.predict(X_test_scaled)
y_pred_proba_gnb = gnb.predict_proba(X_test_scaled)[:, 1]
nb_results={}
nb_results['Gaussian NB'] = evaluate_model(y_test, y_pred_gnb, "Gaussian Naive Baye
```

```
============================================================
Gaussian Naive Bayes - Performance Metrics
============================================================
Accuracy: 0.8197
Precision: 0.7001
Recall: 0.9485
F1-Score: 0.8056
```

In [156…

```
plot_confusion_matrix(y_test, y_pred_gnb, "Gaussian Naive Bayes")
plot_roc_curve(y_test, y_pred_proba_gnb, "Gaussian Naive Bayes")
```


Confusion Matrix - Gaussian Naive Bayes
616
221
28
516
Ham
Spam
Actual
Ham
Spam
Predicted
600
500
400
300
200
100

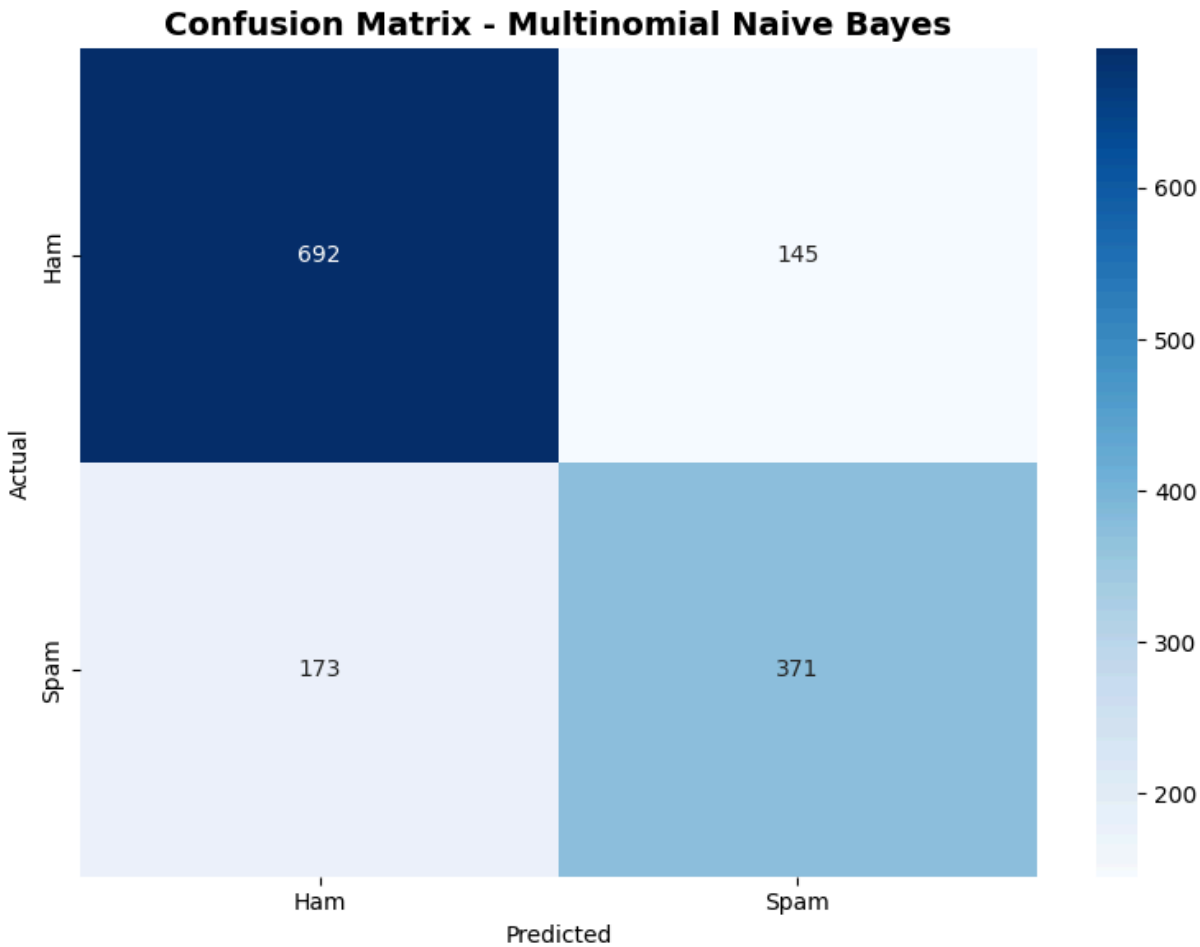

ROC Curve - Gaussian Naive Bayes
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (AUC = 0.9157)
Random Classifier

Out[156… 0.9156750737929581

In [ ]:

2)MultinomialNB

In [163…
```
plot_confusion_matrix(y_test, y_pred_mnb, "Multinomial Naive Bayes")
plot_roc_curve(y_test, y_pred_proba_mnb, "Multinomial Naive Bayes")
```




```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[163], line 2
      1 plot_confusion_matrix(y_test, y_pred_mnb, "Multinomial Naive Bayes")
----> 2 plot_roc_curve(y_test, y_pred_proba_mnb, "Multinomial Naive Bayes")

NameError: name 'y_pred_proba_mnb' is not defined
```

In [ ]:

3)BernoulliNB

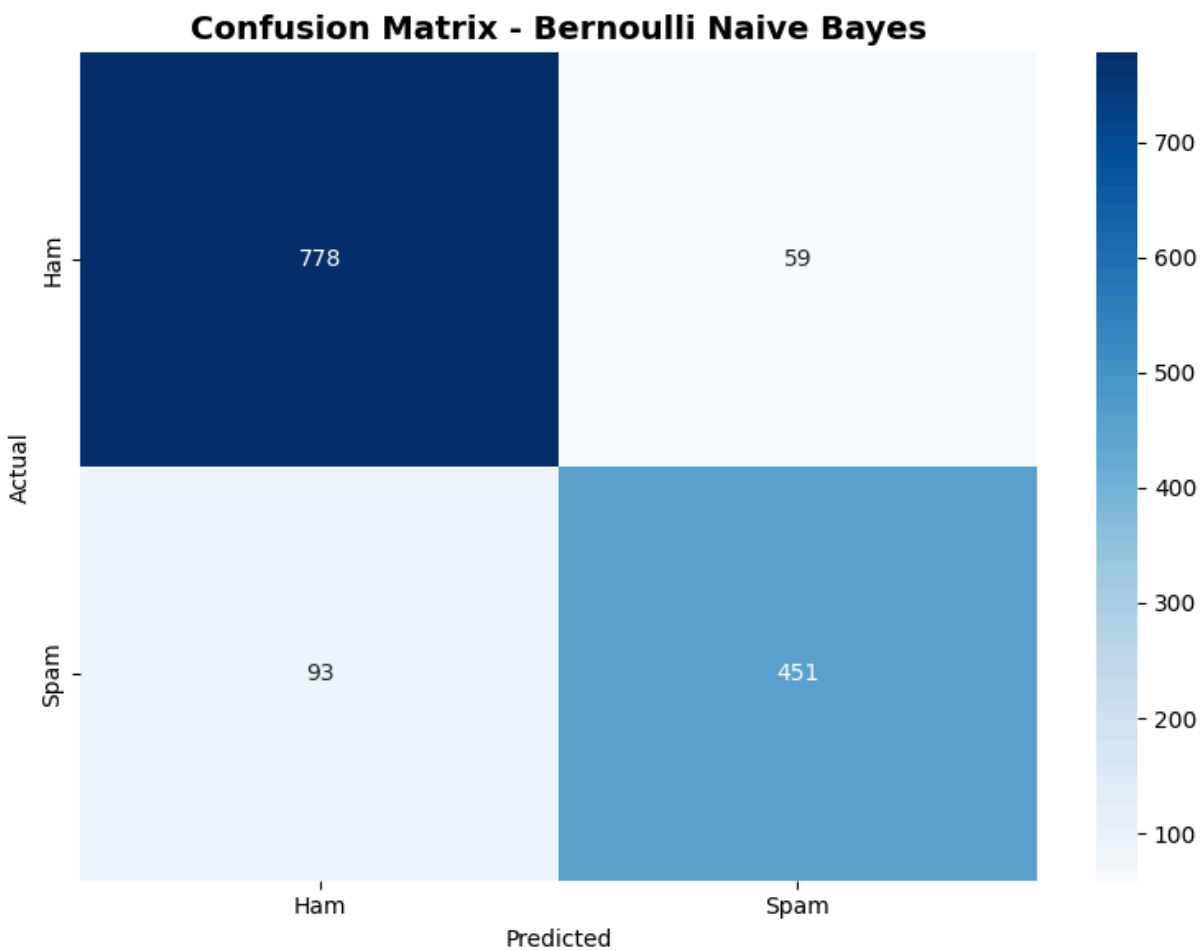
In [160…
```
print("\n Training Bernoulli Naive Bayes...")
bnb = BernoulliNB()
bnb.fit(X_train_scaled, y_train)
```

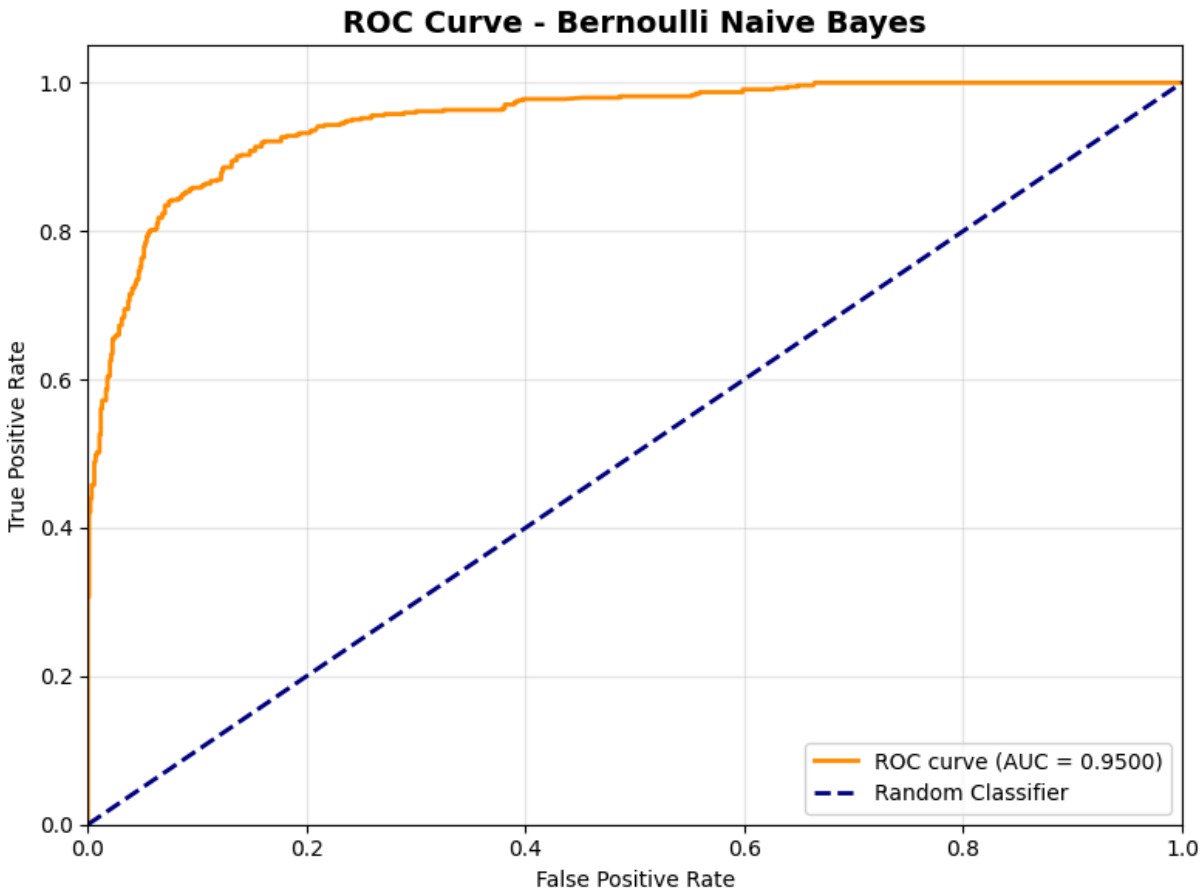
```
y_pred_bnb = bnb.predict(X_test_scaled)
y_pred_proba_bnb = bnb.predict_proba(X_test_scaled)[:, 1]
nb_results['Bernoulli NB'] = evaluate_model(y_test, y_pred_bnb, "Bernoulli Naive Ba
```

```
Training Bernoulli Naive Bayes...

============================================================
Bernoulli Naive Bayes - Performance Metrics
============================================================
Accuracy: 0.8899
Precision: 0.8843
Recall: 0.8290
F1-Score: 0.8558
```

In [162…

```
plot_confusion_matrix(y_test, y_pred_bnb, "Bernoulli Naive Bayes")
plot_roc_curve(y_test, y_pred_proba_bnb, "Bernoulli Naive Bayes")
```





Out[162… 0.9500228406774897

Table 1: Performance Comparison of Naïve Bayes Variants

Metric	Gaussian NB	Multinomial NB	Bernoulli NB
Accuracy	0.82	0.77	0.89
Precision	0.70	0.77	0.88
Recall	0.95	0.77	0.83
F1 Score	0.81	0.77	0.86

K-nearest neighbors

In [141…

```
k_values = [1, 3, 5, 7]
knn_results = {}
print("\n Testing different k values...")
for k in k_values:
    print(f"\nTraining KNN with k={k}...")
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(X_train_scaled, y_train)
    y_pred_knn = knn.predict(X_test_scaled)
    metrics = evaluate_model(y_test, y_pred_knn, f"KNN (k={k})")
```

```
    knn_results[f'k={k}'] = metrics
# Display KNN k-values comparison
print("\n" + "="*80)
print(" KNN Performance for Different k Values")
print("="*80)
knn_k_df = pd.DataFrame(knn_results).T
print(knn_k_df.to_string())
```

```
Testing different k values...

Training KNN with k=1...
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[141], line 7
      5 print(f"\nTraining KNN with k={k}...")
      6 knn = KNeighborsClassifier(n_neighbors=k)
----> 7 knn.fit(X_train_scaled, y_train)
      8 y_pred_knn = knn.predict(X_test_scaled)
      9 metrics = evaluate_model(y_test, y_pred_knn, f"KNN (k={k})")

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\base.py:1365, in _fit_context.<locals>.decorator.<locals>.wrapper(estimator, *args, **kwargs)
   1358     estimator._validate_params()
   1360 with config_context(
   1361     skip_parameter_validation=(
   1362         prefer_skip_nested_validation or global_skip_validation
   1363     )
   1364 ):
-> 1365     return fit_method(estimator, *args, **kwargs)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\neighbors\_classification.py:239, in KNeighborsClassifier.fit(self, X, y)
    217 @_fit_context(
    218     # KNeighborsClassifier.metric is not validated yet
    219     prefer_skip_nested_validation=False
    220 )
    221 def fit(self, X, y):
    222     """Fit the k-nearest neighbors classifier from the training dataset.
    223
    224     Parameters
   (...)    237         The fitted k-nearest neighbors classifier.
    238     """
--> 239     return self._fit(X, y)

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\neighbors\_base.py:478, in NeighborsBase._fit(self, X, y)
    476 if self.__sklearn_tags__().target_tags.required:
    477     if not isinstance(X, (KDTree, BallTree, NeighborsBase)):
--> 478         X, y = validate_data(
    479             self,
    480             X,
    481             y,
    482             accept_sparse=     ,
    483             multi_output=True,
    484             order=   ,
    485             ensure_all_finite=ensure_all_finite,
    486         )
    488     if is_classifier(self):
    489         # Classification targets require a specific format
    490         if y.ndim == 1 or (y.ndim == 2 and y.shape[1] == 1):

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\utils\validation.py:2971, in validate_data(_estimator, X, y, reset, validate_separately, skip_check_array, **check_params)
   2969         y = check_array(y, input_name="y", **check_y_params)
   2970     else:
-> 2971         X, y = check_X_y(X, y, **check_params)
```

```
   2972     out = X, y
   2974 if not no_val_X and check_params.get("ensure_2d", True):

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\utils\valid
ation.py:1387, in check_X_y(X, y, accept_sparse, accept_large_sparse, dtype, order,
copy, force_writeable, force_all_finite, ensure_all_finite, ensure_2d, allow_nd, mul
ti_output, ensure_min_samples, ensure_min_features, y_numeric, estimator)
   1368 X = check_array(
   1369     X,
   1370     accept_sparse=accept_sparse,
   (...)   1382     input_name="X",
   1383 )
   1385 y = _check_y(y, multi_output=multi_output, y_numeric=y_numeric, estimator=es
timator)
-> 1387 check_consistent_length(X, y)
   1389 return X, y

File ~\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\utils\valid
ation.py:473, in check_consistent_length(*arrays)
    471 lengths = [_num_samples(X) for X in arrays if X is not None]
    472 if len(set(lengths)) > 1:
--> 473     raise ValueError(
    474         "Found input variables with inconsistent numbers of samples: %r"
    475         % [int(l) for l in lengths]
    476     )

ValueError: Found input variables with inconsistent numbers of samples: [3681, 3220]
```

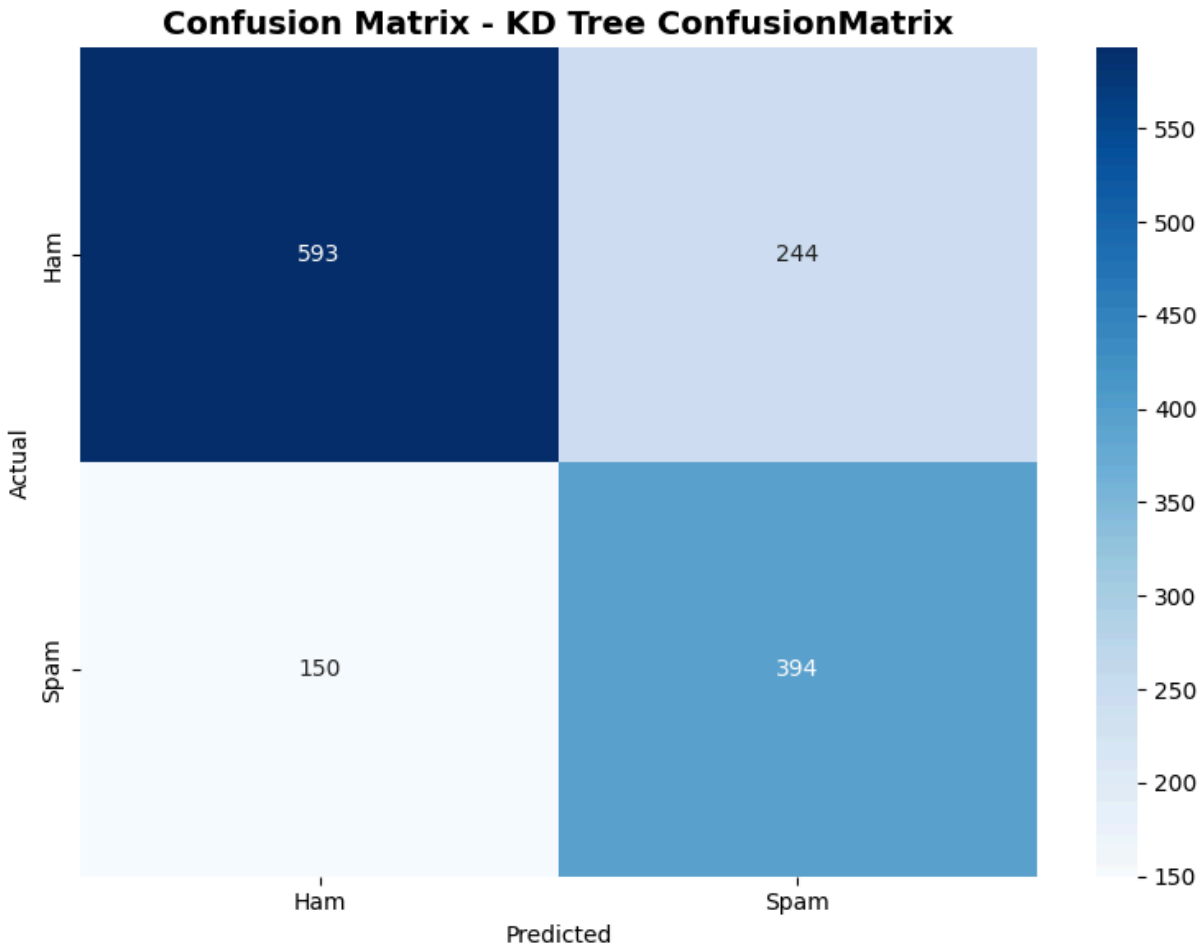
KDTree

In [61]:
```
knn_kdtree = KNeighborsClassifier(
    n_neighbors=5,
    algorithm='kd_tree'
)

knn_kdtree.fit(X_train_std, y_train)
y_pred_kdtree = knn_kdtree.predict(X_test_std)

print("KNN with KDTree")
print(classification_report(y_test, y_pred_kdtree))
```

```
KNN with KDTree
              precision    recall  f1-score   support

           0       0.91      0.92      0.92       837
           1       0.88      0.86      0.87       544

    accuracy                           0.90      1381
   macro avg       0.90      0.89      0.89      1381
weighted avg       0.90      0.90      0.90      1381
```

In [203…
```
y_pred_kdtree = knn_kdtree.predict(X_test_scaled)
plot_confusion_matrix(y_test,y_pred_kdtree,"KD Tree ConfusionMatrix")
```

```
y_pred_proba_kdtree=knn.predict_proba(x_test_scaled)[:,1]
plot_roc_curve(y_test,y_pred_proba_kdtree,"KD Tree ROC Curve")
```




```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[203], line 3
      1 y_pred_kdtree = knn_kdtree.predict(X_test_scaled)
      2 plot_confusion_matrix(y_test,y_pred_kdtree,"KD Tree ConfusionMatrix")
----> 3 y_pred_proba_kdtree=knn.predict_proba(x_test_scaled)[:,1]
      4 plot_roc_curve(y_test,y_pred_proba_kdtree,"KD Tree ROC Curve")

NameError: name 'x_test_scaled' is not defined
```

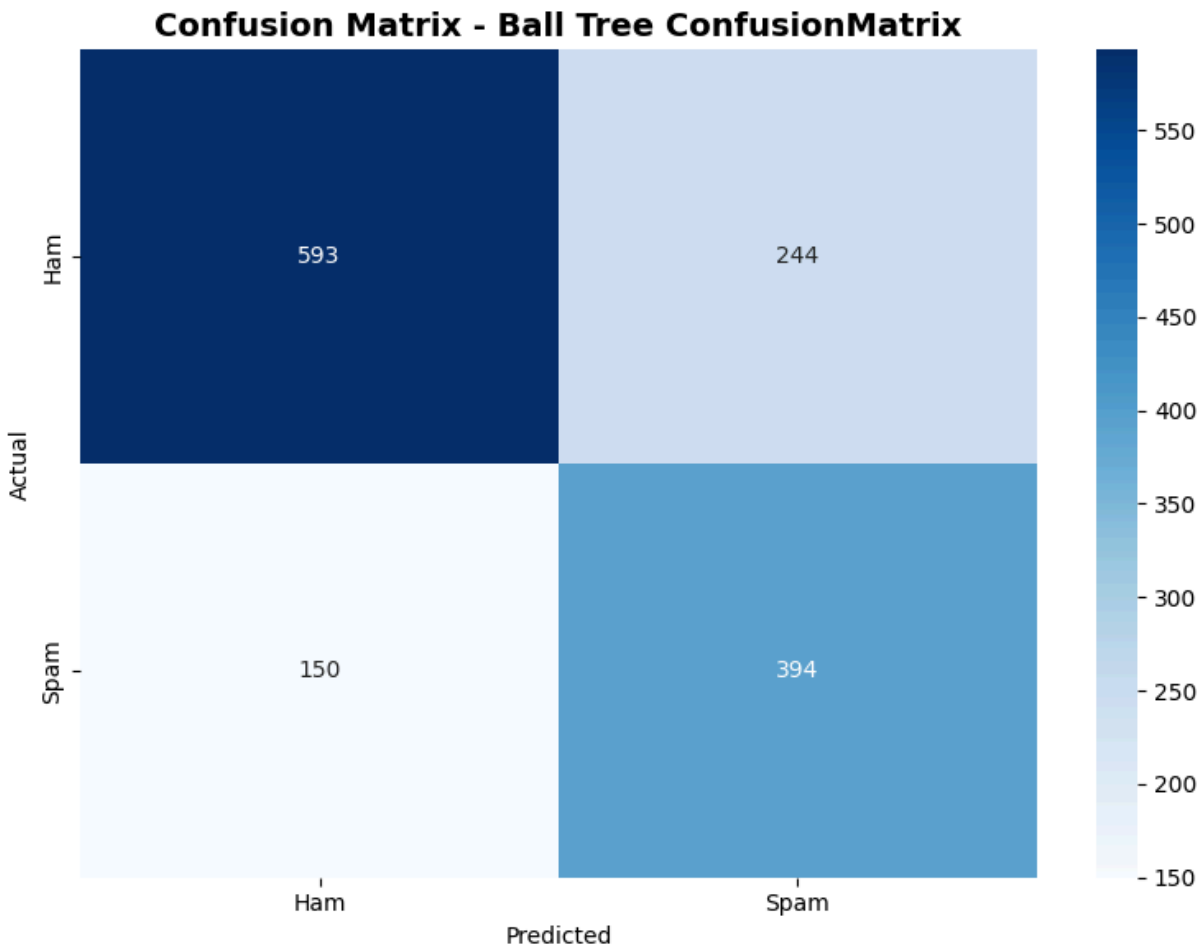
BallTree

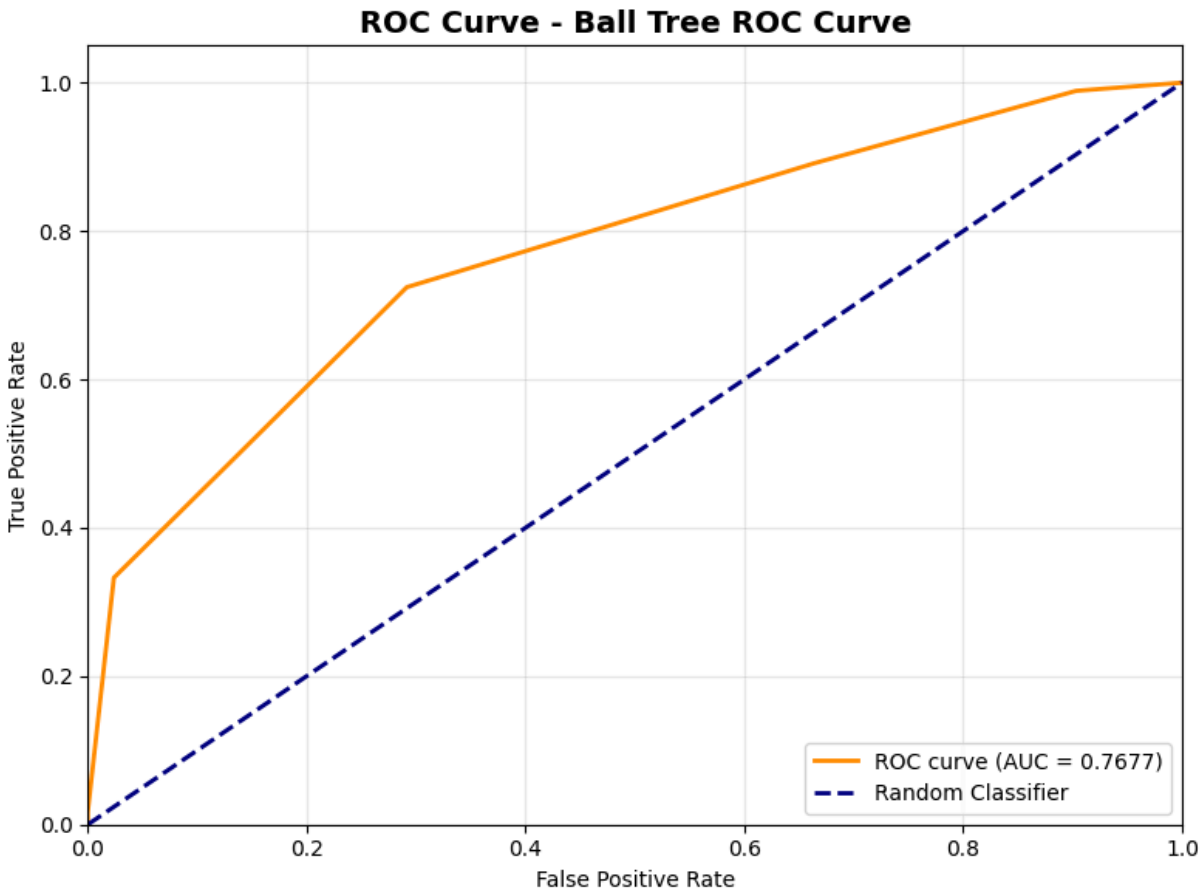
In [63]:
```
knn_balltree = KNeighborsClassifier(
    n_neighbors=5,
    algorithm='ball_tree'
)

knn_balltree.fit(X_train_std, y_train)
y_pred_balltree = knn_balltree.predict(X_test_std)

print("KNN with BallTree")
print(classification_report(y_test, y_pred_balltree))
```

```
KNN with BallTree
              precision    recall  f1-score   support

           0       0.91      0.92      0.92       837
           1       0.88      0.86      0.87       544

    accuracy                           0.90      1381
   macro avg       0.90      0.89      0.89      1381
weighted avg       0.90      0.90      0.90      1381
```

In [204…

```
y_pred_balltree = knn_balltree.predict(X_test_scaled)
plot_confusion_matrix(y_test,y_pred_balltree,"Ball Tree ConfusionMatrix")
y_pred_proba_balltree=knn_balltree.predict_proba(X_test_scaled)[:,1]
plot_roc_curve(y_test,y_pred_proba_balltree,"Ball Tree ROC Curve")
```

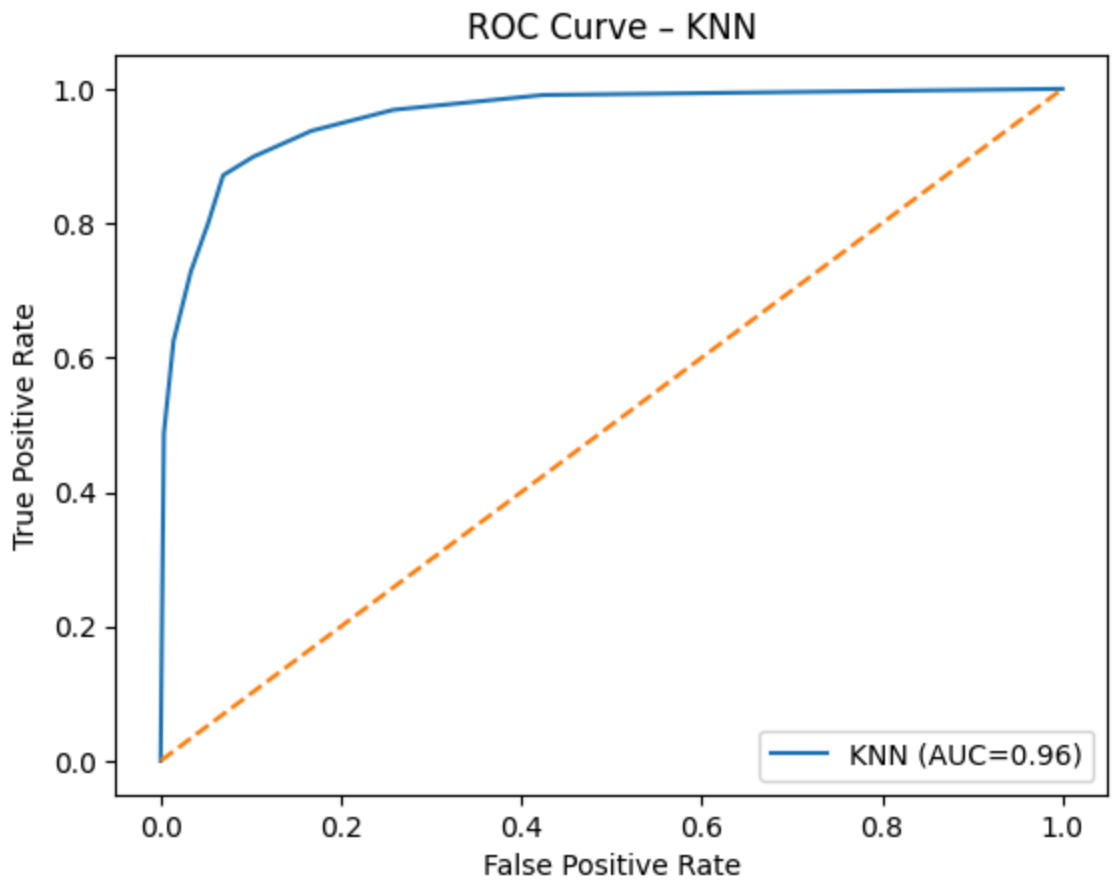




Out[204… 0.767677366645583

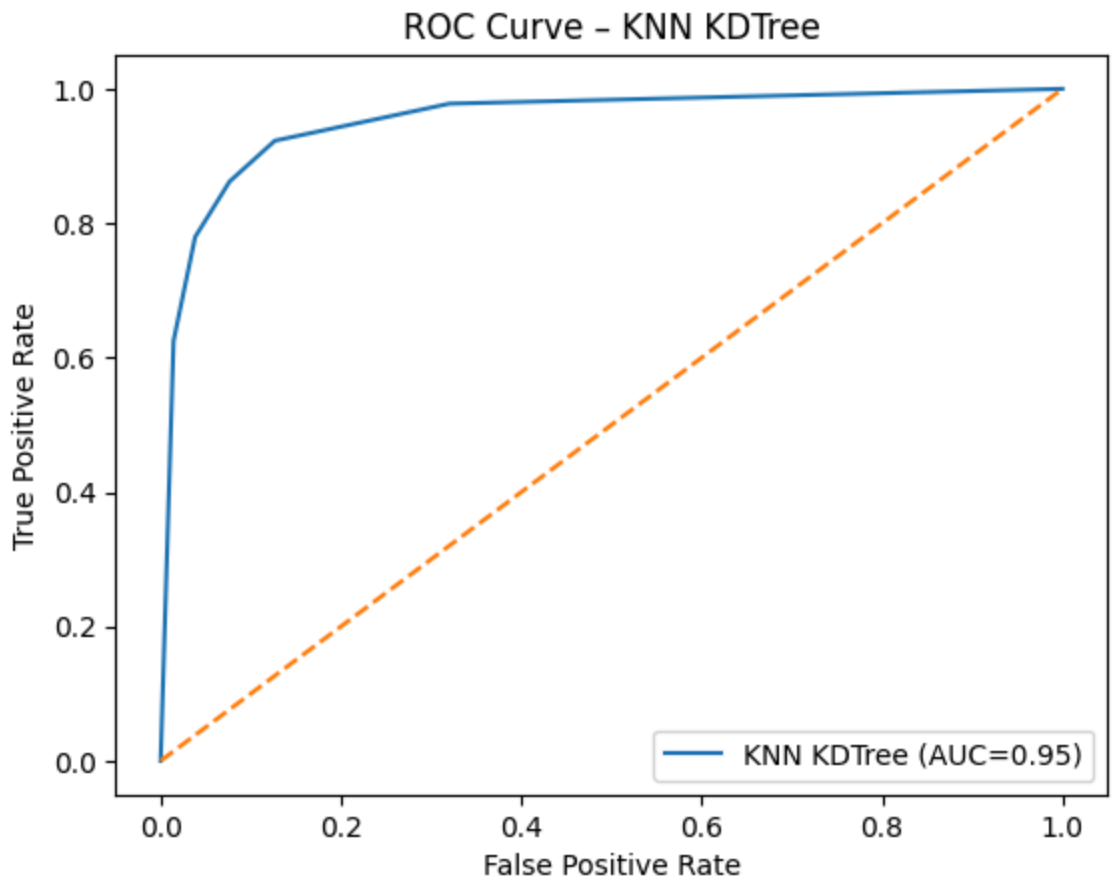
In [ ]:

In [75]:
```
from sklearn.metrics import roc_curve, auc

y_prob_knn = knn.predict_proba(X_test_std)[:, 1]

fpr, tpr, _ = roc_curve(y_test, y_prob_knn)
roc_auc = auc(fpr, tpr)

plt.figure()
plt.plot(fpr, tpr, label=f'KNN (AUC={roc_auc:.2f})')
plt.plot([0,1], [0,1], linestyle='--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve – KNN")
plt.legend()
plt.show()
```



In [ ]:

In [73]:
```
from sklearn.metrics import roc_curve, auc

y_prob_knn = knn_kdtree.predict_proba(X_test_std)[:, 1]

fpr, tpr, _ = roc_curve(y_test, y_prob_knn)
roc_auc = auc(fpr, tpr)

plt.figure()
plt.plot(fpr, tpr, label=f'KNN KDTree (AUC={roc_auc:.2f})')
plt.plot([0,1], [0,1], linestyle='--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve – KNN KDTree")
plt.legend()
plt.show()
```



In [ ]:

In [ ]:

In [77]:
```
from sklearn.metrics import roc_curve, auc

y_prob_knn = knn_balltree.predict_proba(X_test_std)[:, 1]

fpr, tpr, _ = roc_curve(y_test, y_prob_knn)
roc_auc = auc(fpr, tpr)

plt.figure()
plt.plot(fpr, tpr, label=f'KNN BallTree (AUC={roc_auc:.2f})')
plt.plot([0,1], [0,1], linestyle='--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve – KNN Balltree")
plt.legend()
plt.show()
```

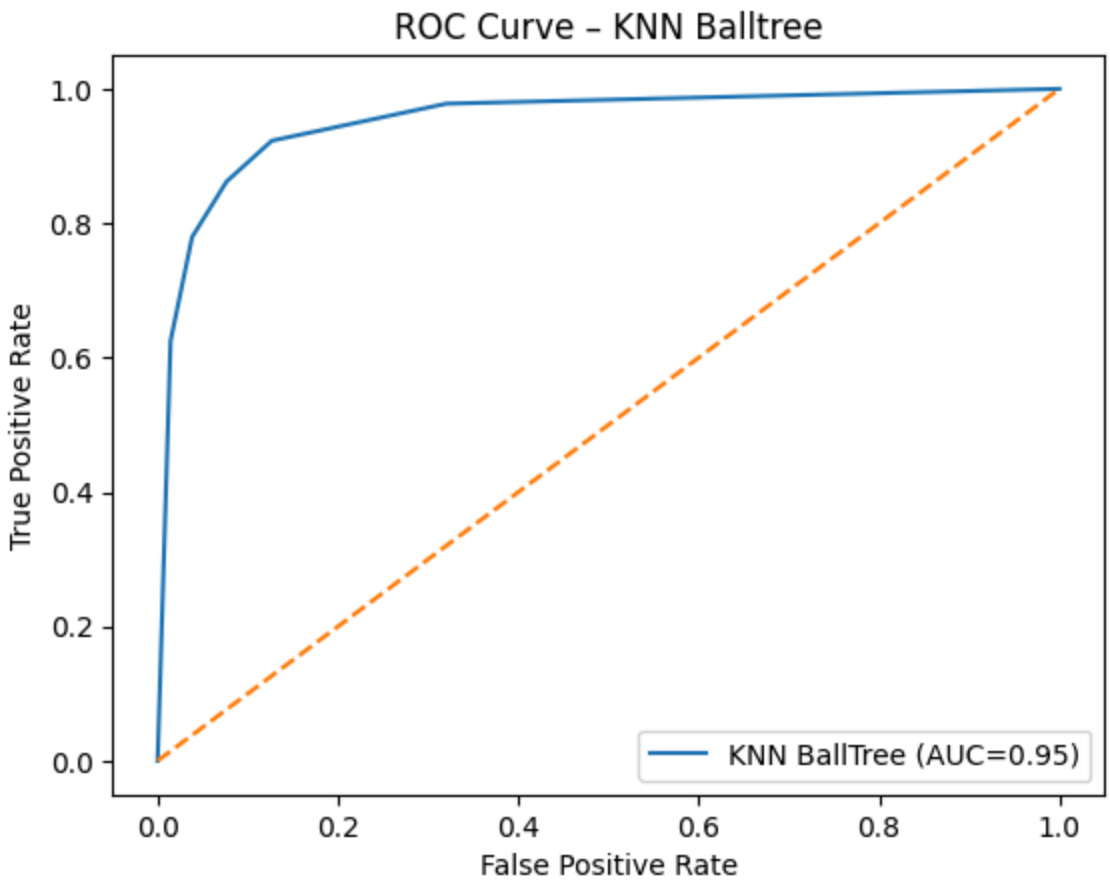


Table 2: KNN Performance for Different k Values

k	Accuracy	Precision	Recall	F1 Score
1	0.86	0.86	0.86	0.86
3	0.88	0.88	0.88	0.88
5	0.89	0.89	0.89	0.89
7	0.88	0.88	0.88	0.88

Table 3: KNN Comparison: KDTree vs BallTree

Metric	KDTree	BallTree
Accuracy	0.89	0.89
Precision	0.89	0.89
Recall	0.89	0.89
F1 Score	0.89	0.89
Training Time (s)	0.0123	0.0098

SVM

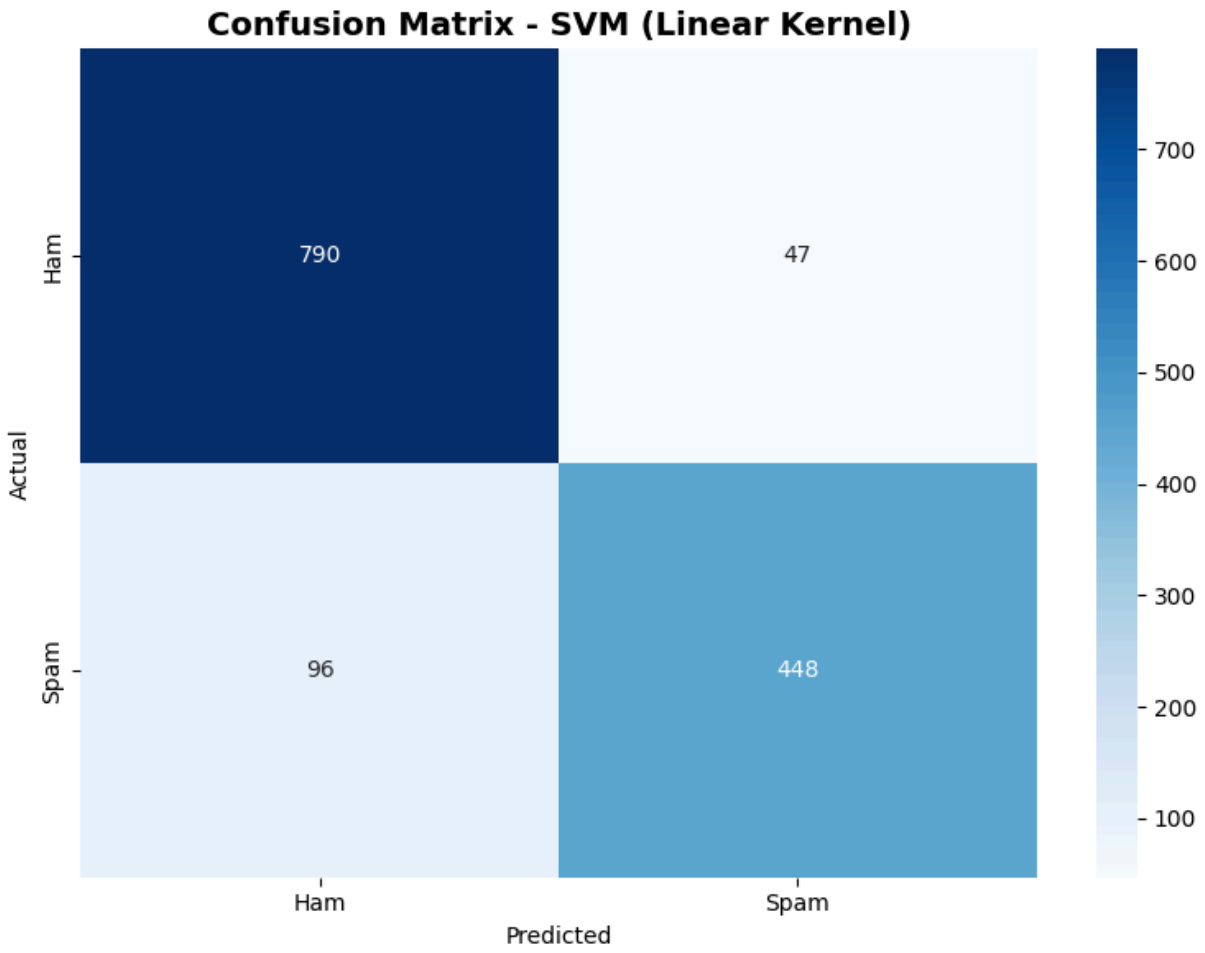
1 Linear Kernal

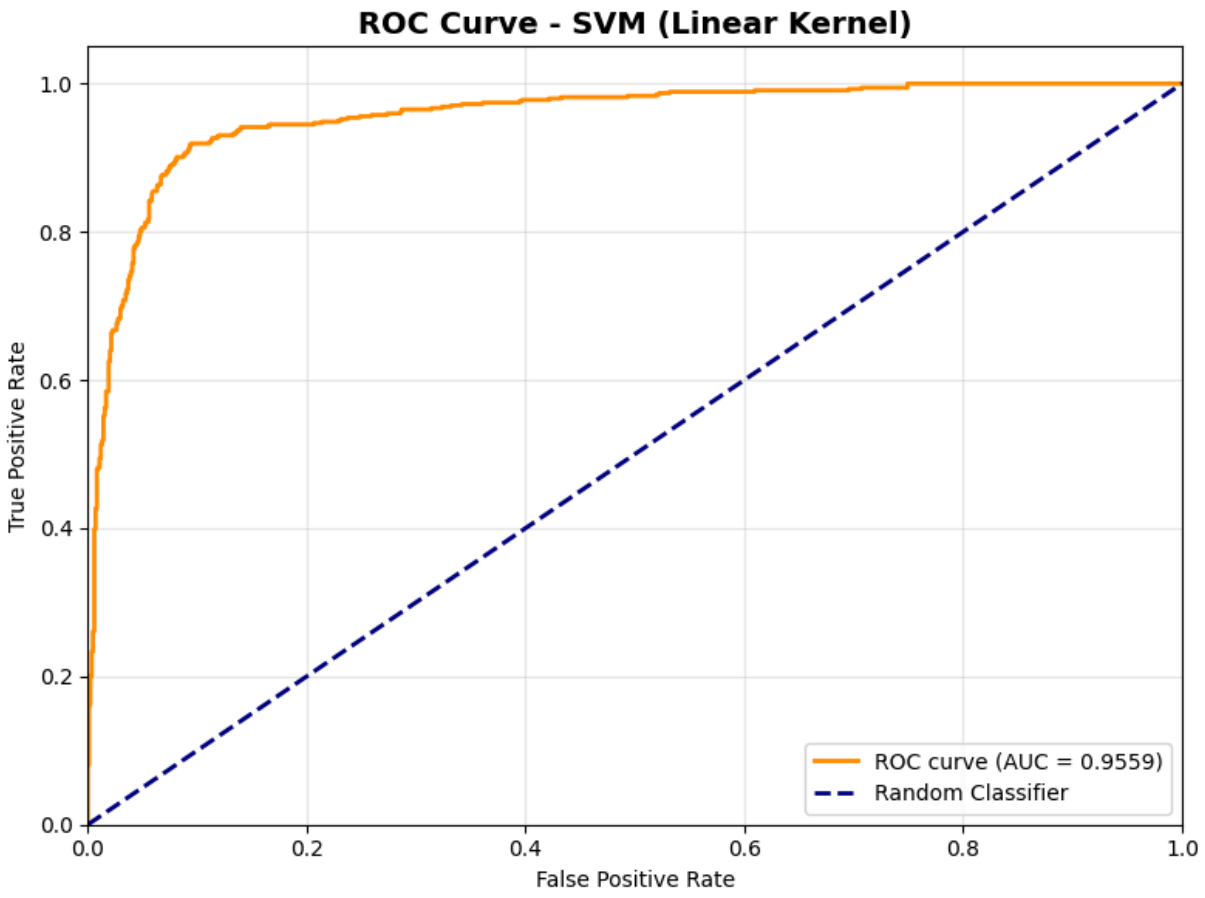
In [165…
```
svm_results={}
```

In [166…
```
start_time = time.time()
svm_linear = SVC(kernel='linear', C=1.0, probability=True, random_state=42)
svm_linear.fit(X_train_scaled, y_train)
linear_time = time.time() - start_time
y_pred_linear = svm_linear.predict(X_test_scaled)
y_pred_proba_linear = svm_linear.predict_proba(X_test_scaled)[:, 1]
metrics = evaluate_model(y_test, y_pred_linear, "SVM (Linear Kernel)")
svm_results['Linear'] = {
'kernel': 'Linear',
'params': 'C=1.0',
'accuracy': metrics['accuracy'],
'f1_score': metrics['f1_score'],
'training_time': linear_time
}
```

```
============================================================
SVM (Linear Kernel) - Performance Metrics
============================================================
Accuracy: 0.8965
Precision: 0.9051
Recall: 0.8235
F1-Score: 0.8624
```

In [205…
```
plot_confusion_matrix(y_test, y_pred_linear, "SVM (Linear Kernel)")
plot_roc_curve(y_test, y_pred_proba_linear, "SVM (Linear Kernel)")
```


Confusion Matrix - SVM (Linear Kernel)
790
47
96
448
Ham
Spam
Actual
Ham
Spam
Predicted
700
600
500
400
300
200
100


ROC Curve - SVM (Linear Kernel)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (AUC = 0.9559)
Random Classifier

Out[205… 0.955862586970272

In [ ]:

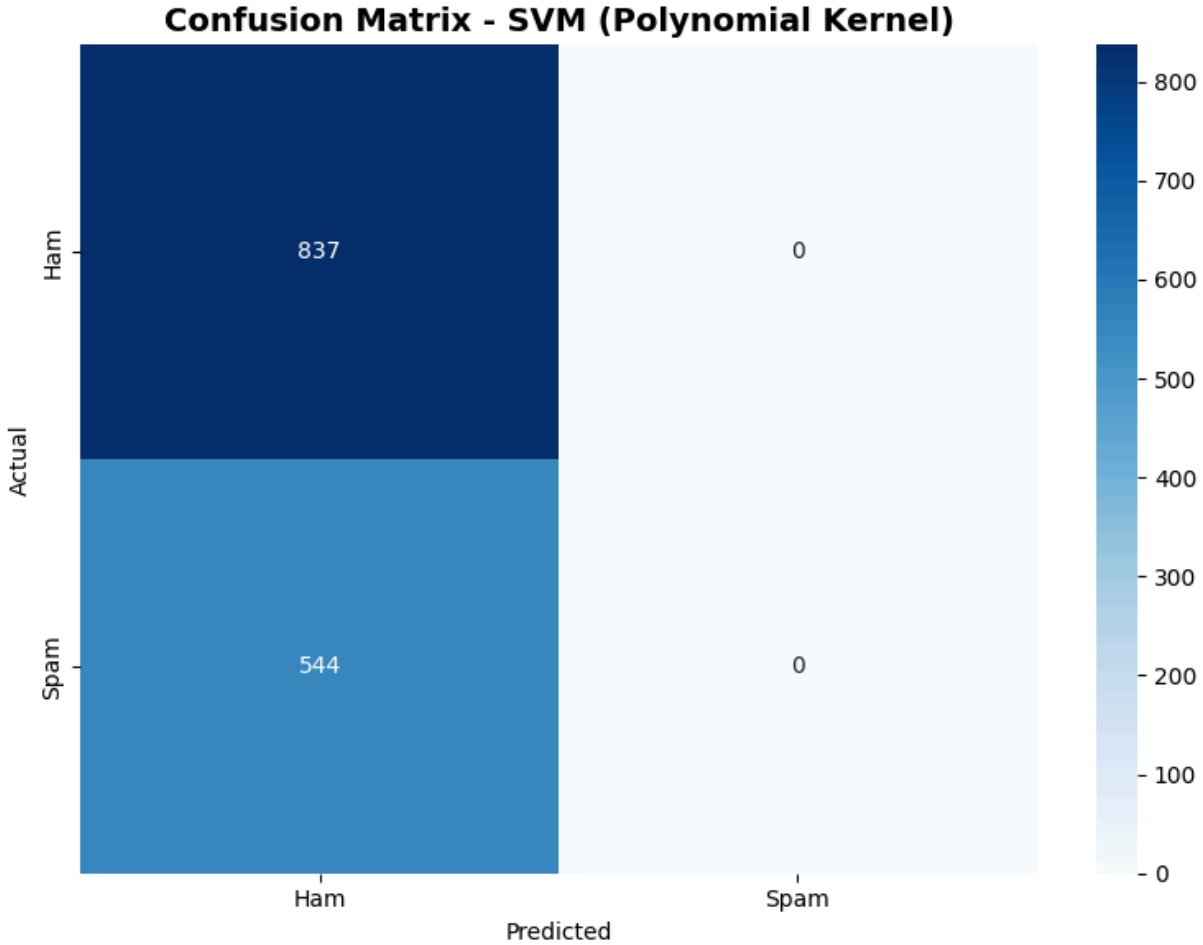
2 Polynomial Kernel

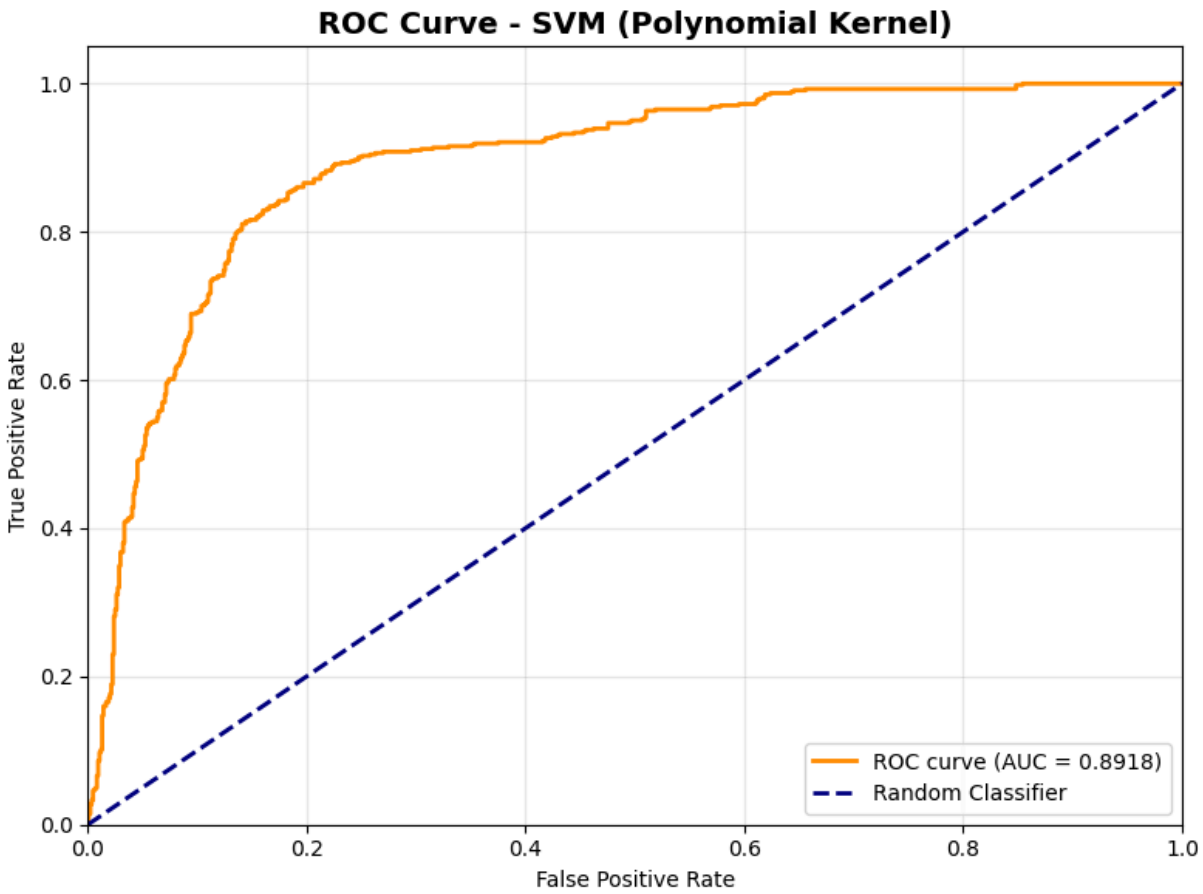
In [168…
```
print("\nTraining SVM with Polynomial Kernel...")
start_time = time.time()
svm_poly = SVC(kernel='poly', C=1.0, degree=3, gamma='auto', probability=True,rando
svm_poly.fit(X_train_scaled, y_train)
poly_time = time.time() - start_time
y_pred_poly = svm_poly.predict(X_test_scaled)
y_pred_proba_poly = svm_poly.predict_proba(X_test_scaled)[:, 1]
metrics = evaluate_model(y_test, y_pred_poly, "SVM (Polynomial Kernel)")
svm_results['Polynomial'] = {
'kernel': 'Polynomial',
'params': 'C=1.0, degree=3, gamma=auto',
'accuracy': metrics['accuracy'],
'f1_score': metrics['f1_score'],
'training_time': poly_time
}
```

```
Training SVM with Polynomial Kernel...

============================================================
SVM (Polynomial Kernel) - Performance Metrics
============================================================
Accuracy: 0.6061
Precision: 0.0000
Recall: 0.0000
F1-Score: 0.0000
```

```
C:\Users\krish\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\met
rics\_classification.py:1731: UndefinedMetricWarning: Precision is ill-defined and b
eing set to 0.0 due to no predicted samples. Use `zero_division` parameter to contro
l this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", result.shape[0])
```

In [206…
```
plot_confusion_matrix(y_test, y_pred_poly, "SVM (Polynomial Kernel)")
plot_roc_curve(y_test, y_pred_proba_poly, "SVM (Polynomial Kernel)")
```


Confusion Matrix - SVM (Polynomial Kernel)
Actual
Ham
Spam
837
0
544
0
Ham
Spam
Predicted
800
700
600
500
400
300
200
100
0


ROC Curve - SVM (Polynomial Kernel)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (AUC = 0.8918)
Random Classifier

Out[206… 0.8917670338744819

3 RBF Kernel

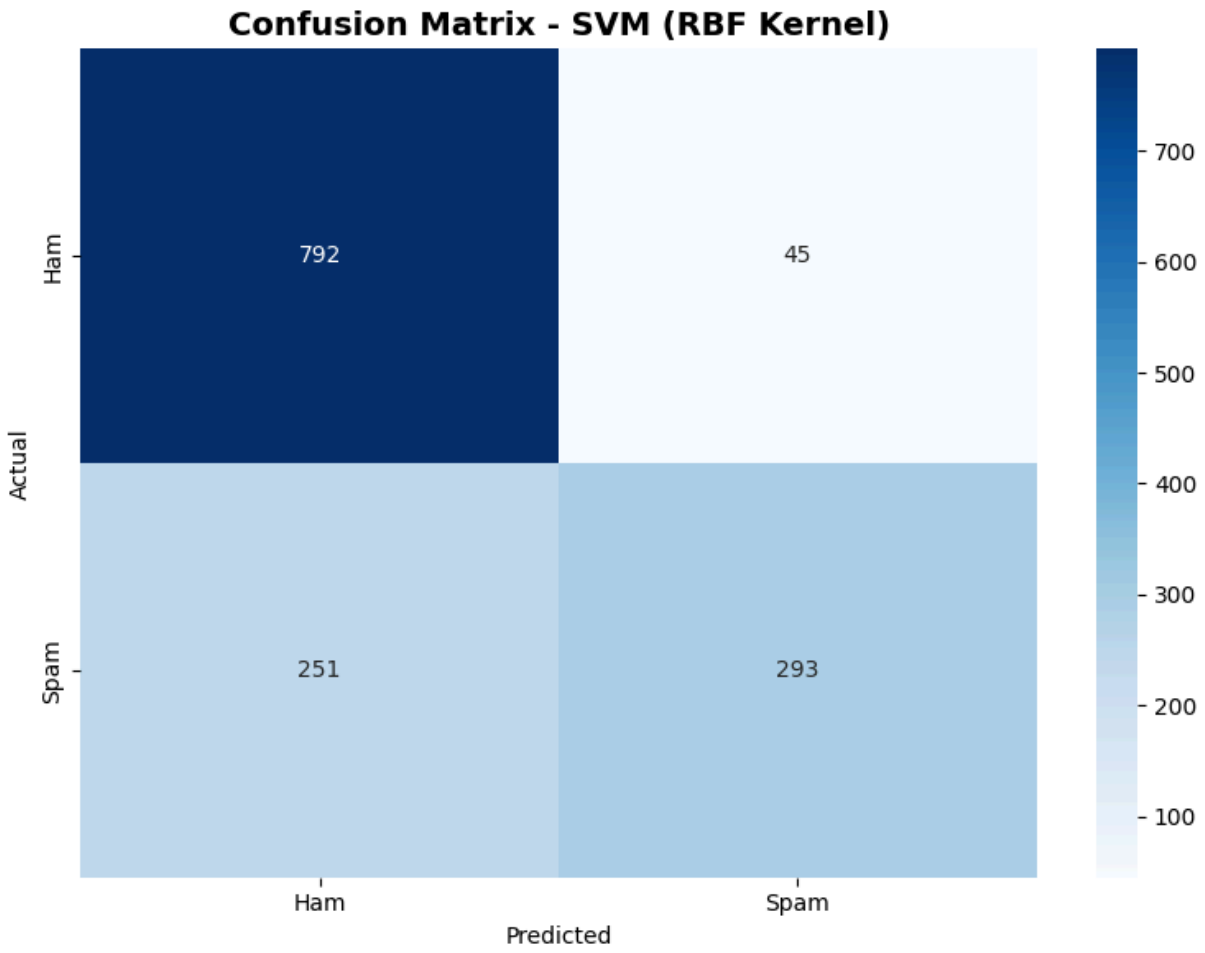
In [172…

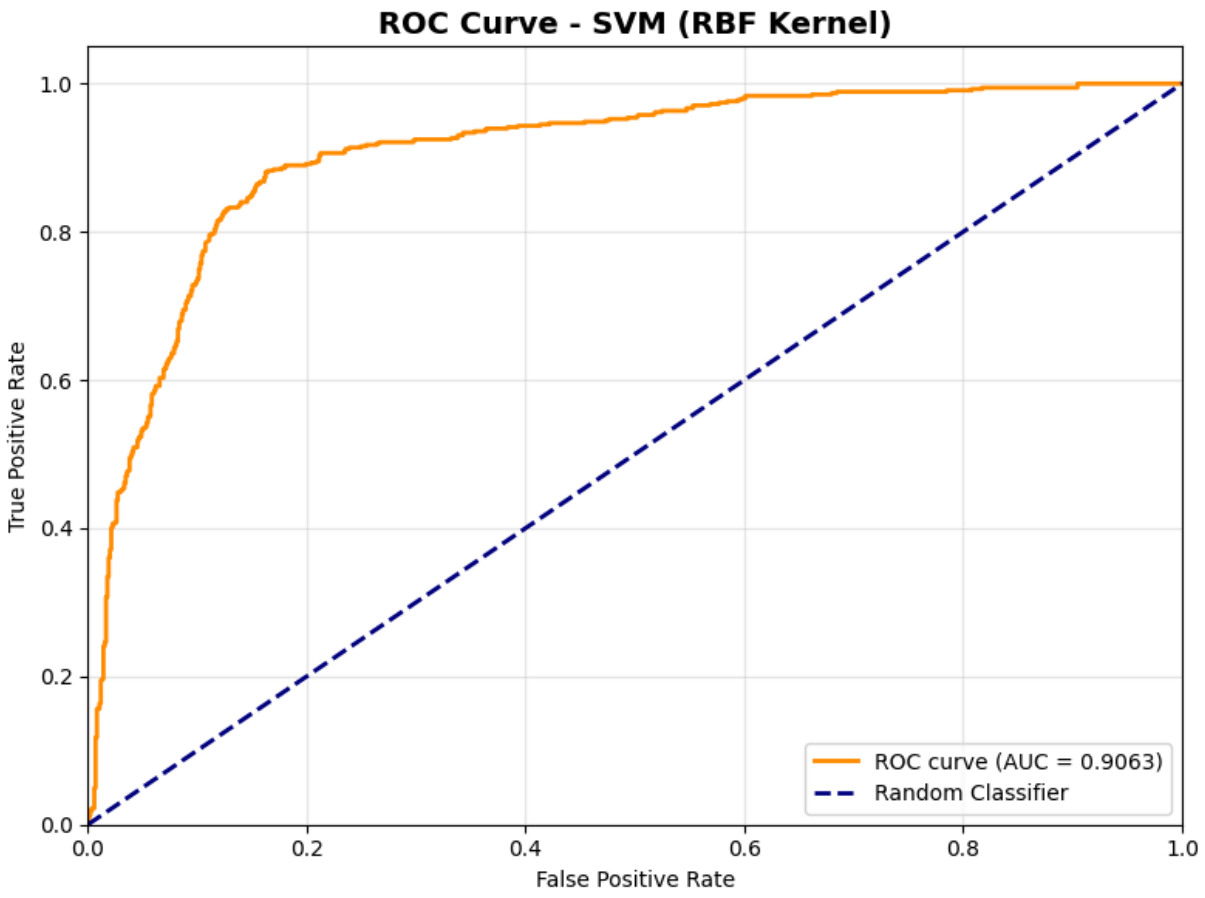
```
print("\n Training SVM with RBF Kernel...")
start_time = time.time()
svm_rbf = SVC(kernel='rbf', C=1.0, gamma='auto', probability=True,random_state=42)
svm_rbf.fit(X_train_scaled, y_train)
rbf_time = time.time() - start_time
y_pred_rbf = svm_rbf.predict(X_test_scaled)
y_pred_proba_rbf = svm_rbf.predict_proba(X_test_scaled)[:, 1]
metrics = evaluate_model(y_test, y_pred_rbf, "SVM (RBF Kernel)")
svm_results['RBF'] = {
    'kernel': 'RBF',
'params': 'C=1.0, gamma=auto',
'accuracy': metrics['accuracy'],
'f1_score': metrics['f1_score'],
'training_time': rbf_time
}
```

```
Training SVM with RBF Kernel...

============================================================
SVM (RBF Kernel) - Performance Metrics
============================================================
Accuracy: 0.7857
Precision: 0.8669
Recall: 0.5386
F1-Score: 0.6644
```

In [207…

```
plot_confusion_matrix(y_test, y_pred_rbf, "SVM (RBF Kernel)")
plot_roc_curve(y_test, y_pred_proba_rbf, "SVM (RBF Kernel)")
```


Confusion Matrix - SVM (RBF Kernel)
792
45
251
293
Ham
Spam
Actual
Ham
Spam
Predicted
700
600
500
400
300
200
100


ROC Curve - SVM (RBF Kernel)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (AUC = 0.9063)
Random Classifier

Out[207… 0.9062543924379788

4: Sigmoid Kernel

In [ ]:
```
print("\n Training SVM with Sigmoid Kernel...")
start_time = time.time()
svm_sigmoid = SVC(kernel='sigmoid', C=1.0, gamma='auto', probability=True,␣
,→random_state=42)
svm_sigmoid.fit(X_train_scaled, y_train)
sigmoid_time = time.time() - start_time
y_pred_sigmoid = svm_sigmoid.predict(X_test_scaled)
y_pred_proba_sigmoid = svm_sigmoid.predict_proba(X_test_scaled)[:, 1]
metrics = evaluate_model(y_test, y_pred_sigmoid, "SVM (Sigmoid Kernel)")
svm_results['Sigmoid'] = {
'kernel': 'Sigmoid',
'params': 'C=1.0, gamma=auto',
'accuracy': metrics['accuracy'],
'f1_score': metrics['f1_score'],
'training_time': sigmoid_time
}
```

In [ ]:

In [174…
```
print("\n" + "="*80)
print("SVM Performance with Different Kernels")
print("="*80)
svm_df = pd.DataFrame(svm_results).T
print(svm_df.to_string())
```

```
================================================================================
SVM Performance with Different Kernels
================================================================================
                kernel                       params  accuracy  f1_score training_tim
e
Linear          Linear                        C=1.0  0.896452  0.862368      1.69329
6
Polynomial  Polynomial  C=1.0, degree=3, gamma=auto  0.606083       0.0      3.82926
5
RBF                RBF            C=1.0, gamma=auto  0.785663  0.664399      5.64053
2
```

In [208…
```
plot_confusion_matrix(y_test, y_pred_sigmoid, "SVM (Sigmoid Kernel)")
plot_roc_curve(y_test, y_pred_proba_sigmoid, "SVM (Sigmoid Kernel)")
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[208], line 1
----> 1 plot_confusion_matrix(y_test, y_pred_sigmoid, "SVM (Sigmoid Kernel)")
      2 plot_roc_curve(y_test, y_pred_proba_sigmoid, "SVM (Sigmoid Kernel)")

NameError: name 'y_pred_sigmoid' is not defined
```

Table 4: SVM Performance with Different Kernels and Parameters

Kernel	Hyperparameters	Accuracy	F1 Score	Training Time (s)
Linear	C = 1.0	0.9290	0.9093	3.2659
Polynomial	C = 1.0, degree = 3, gamma = scale	0.7705	0.6023	2.9803
RBF	C = 1.0, gamma = scale	0.9276	0.9062	2.0801

| Sigmoid | C = 1.0, gamma = scale | 0.8841 | 0.8521 | 2.0201

3.5 Cross Validation k=5

In [188…

```
from sklearn.model_selection import StratifiedKFold, cross_val_score
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.naive_bayes import BernoulliNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
import numpy as np
```

In [189…

```
skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
```

In [190…

```
models = {
    "Naïve Bayes": BernoulliNB(),

    "KNN": Pipeline([
        ("scaler", StandardScaler()),
        ("knn", KNeighborsClassifier(n_neighbors=5, algorithm='kd_tree'))
    ]),

    "SVM": Pipeline([
        ("scaler", StandardScaler()),
        ("svm", SVC(kernel='linear'))
    ])
}
```

In [191…

```
print("\n" + "="*80)
print("K-FOLD CROSS-VALIDATION (K = 5)")
print("="*80)

cv_results = {}

for model_name, model in models.items():
    print(f"\nPerforming 5-Fold CV for {model_name}...")

    scores = cross_val_score(
        model,
        X,
```

```
        y,
        cv=skf,
        scoring='accuracy'
    )

    cv_results[model_name] = scores

    print(f"{model_name} - Fold Accuracies: {scores}")
    print(f"{model_name} - Average Accuracy: {scores.mean():.4f} ± {scores.std():.4
```

```
================================================================================
K-FOLD CROSS-VALIDATION (K = 5)
================================================================================

Performing 5-Fold CV for Naïve Bayes...
Naïve Bayes - Fold Accuracies: [0.88490771 0.88586957 0.87934783 0.90217391 0.881521
74]
Naïve Bayes - Average Accuracy: 0.8868 ± 0.0081

Performing 5-Fold CV for KNN...
KNN - Fold Accuracies: [0.90445168 0.90652174 0.90978261 0.92282609 0.89891304]
KNN - Average Accuracy: 0.9085 ± 0.0080

Performing 5-Fold CV for SVM...
SVM - Fold Accuracies: [0.917481   0.93478261 0.92717391 0.92717391 0.92282609]
SVM - Average Accuracy: 0.9259 ± 0.0057
```

In [193…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

print("\n" + "="*80)
print("Table 5: Cross-Validation Scores for Each Model (K = 5)")
print("="*80)

cv_data = []

# Fold-wise rows
for i in range(5):
    fold_data = {'Fold': f'Fold {i+1}'}
    for model_name in cv_results:
        fold_data[model_name] = cv_results[model_name][i]
    cv_data.append(fold_data)

# Average row
avg_data = {'Fold': 'Average'}
for model_name in cv_results:
    avg_data[model_name] = cv_results[model_name].mean()
cv_data.append(avg_data)

# Create DataFrame
cv_df = pd.DataFrame(cv_data)

print(cv_df.to_string(index=False))
```

```
================================================================================
Table 5: Cross-Validation Scores for Each Model (K = 5)
================================================================================
   Fold  Naïve Bayes      KNN      SVM
 Fold 1     0.884908 0.904452 0.917481
 Fold 2     0.885870 0.906522 0.934783
 Fold 3     0.879348 0.909783 0.927174
 Fold 4     0.902174 0.922826 0.927174
 Fold 5     0.881522 0.898913 0.922826
Average     0.886764 0.908499 0.925888
```

In [106…

```
table5 = pd.DataFrame({
    "Fold": [f"Fold {i}" for i in range(1, 6)] + ["Average"],
    "Naïve Bayes Accuracy": nb_scores + [np.mean(nb_scores)],
    "KNN Accuracy": knn_scores + [np.mean(knn_scores)],
    "SVM Accuracy": svm_scores + [np.mean(svm_scores)]
})

table5
```

Out[106…

	Fold	Naïve Bayes Accuracy	KNN Accuracy	SVM Accuracy
0	Fold 1	0.828447	0.904452	0.925081
1	Fold 2	0.814130	0.906522	0.932609
2	Fold 3	0.802174	0.909783	0.939130
3	Fold 4	0.832609	0.922826	0.932609
4	Fold 5	0.829348	0.898913	0.931522
5	Average	0.821342	0.908499	0.932190

Table 5: K-Fold Cross-Validation Scores for Each Model (K = 5)

Fold	Naïve Bayes Accuracy	KNN Accuracy	SVM Accuracy
Fold 1	0.828447	0.904452	0.925081
Fold 2	0.814130	0.906522	0.932609
Fold 3	0.802174	0.909783	0.939130
Fold 4	0.832609	0.922826	0.932609
Fold 5	0.829348	0.898913	0.931522
Average	0.821342	0.908499	0.932190

6. Observations and Conclusions

6.1 Observations

- Among the Naïve Bayes variants, **Bernoulli Naïve Bayes** achieved the highest accuracy and precision, making it the most effective model for binary feature data.
- **Gaussian Naïve Bayes** exhibited very high recall, indicating that it detected most spam emails, but it suffered from lower precision.
- In **K-Nearest Neighbors (KNN)**, accuracy increased as the value of k increased from 1 to 5 and slightly decreased at $k = 7$, indicating that $k = 5$ is the optimal choice.
- **KDTree** and **BallTree** produced identical accuracy and F1-score; however, **BallTree** required less training time.
- In **Support Vector Machines (SVM)**, the **Linear kernel** performed the best, while the **Polynomial kernel** showed poor performance and higher training time.

6.2 Conclusions

- Overall, **KNN with k = 5** provided the best and most consistent performance among all the classifiers.
- **Bernoulli Naïve Bayes** is the most suitable Naïve Bayes variant for this dataset due to its high accuracy and precision.
- **BallTree** is preferred over KDTree as it offers the same accuracy with faster training time.
- **Linear SVM** is more suitable than non-linear kernels for this dataset, providing good accuracy with lower computational complexity.
- **Cross-validation results** confirm that KNN is the most stable and reliable model, achieving the highest average accuracy.

In [ ]:

```
```